

Models, calibration and inference

The construction of a volatility surface from option quotes, and its evolution between observations.

Market inputs

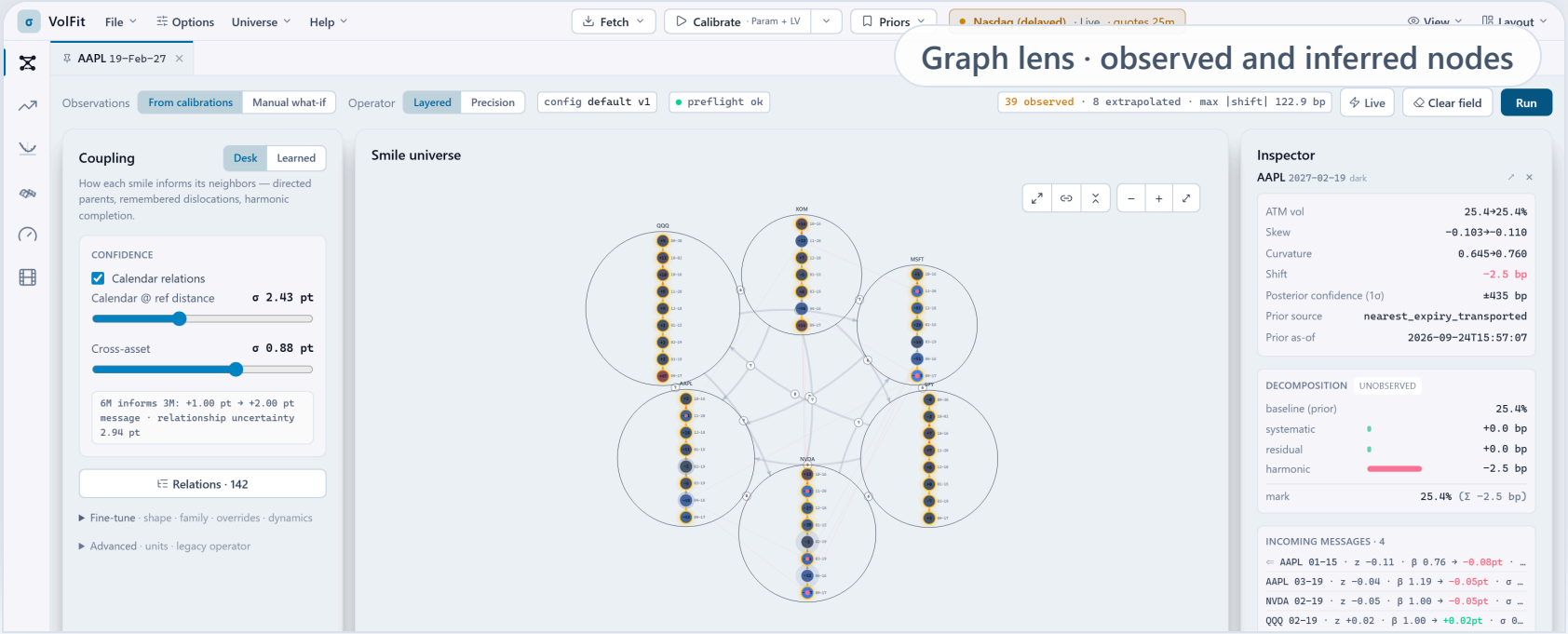
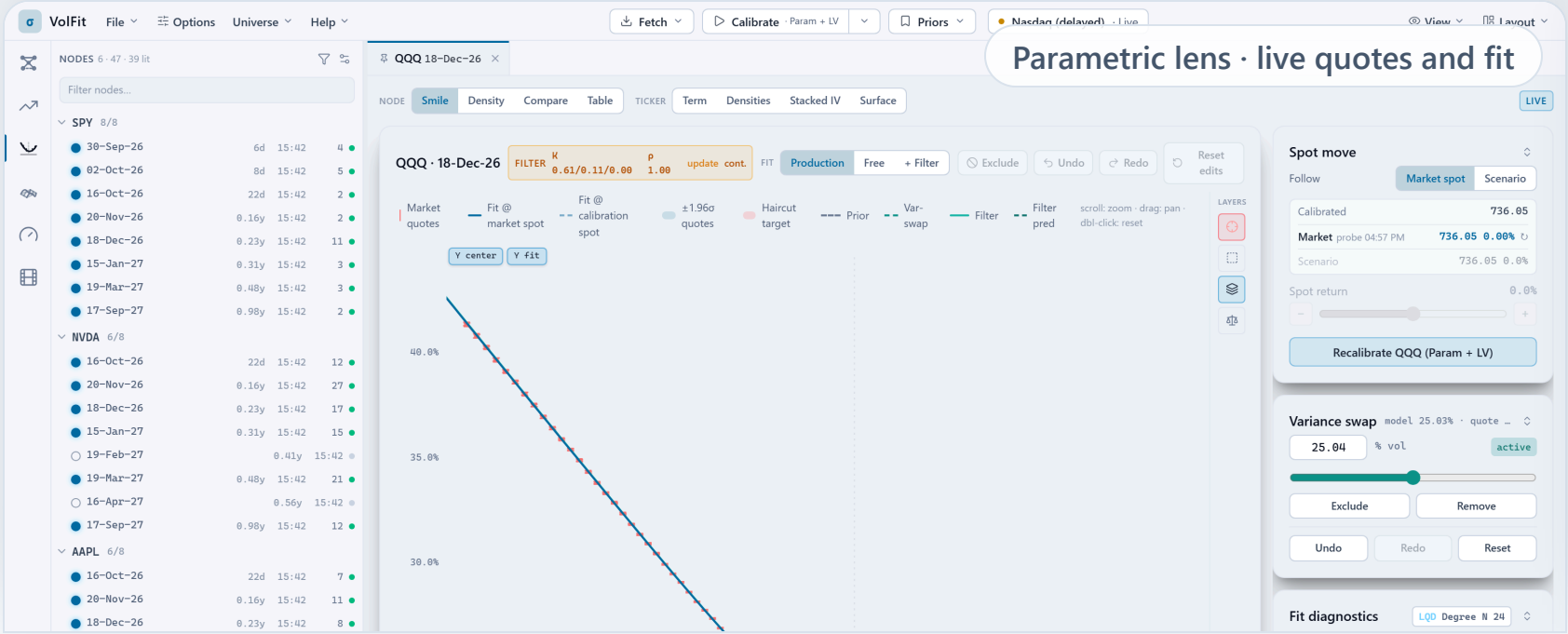
Forwards, dividends, exercise style, quote timestamps and bid–ask information.

Calibration

Smile models, local volatility, the objective, tail assumptions and consistency across maturities.

Subsequent updates

Event time, spot transport, prior persistence, temporal filtering and graph inference.



Objects and calculation order

A node is one underlying and one expiry. A surface couples the nodes belonging to an underlying.

STAGE	INPUT	OUTPUT
Prepare	Quotes, exercise style, spot, carry	European prices, F, D and total variance
Fit	Prepared quotes + objective + model	A smile, or a local-variance surface
Constrain	Adjacent expiries and tail policy	Calendar diagnostics and a repaired surface
Update	Spot move and saved state	Transported fit, prior or filtered estimate
Infer	Observed handle changes and relations	Dark-node handles, smile and uncertainty

NOTATION AND UNITS

$x = K/F$; $k = \log(K/F)$; $c = C/(DF)$; $w = \sigma^2\tau$. Calendar time t drives carry and exercise. The variance clock τ defines the volatility reading. One vol bp is 0.0001 of absolute volatility.

IN THE APP

Open one node on the Parametric lens and read the Fit diagnostics card: the three handles, the RMS on the smile and on the surface.

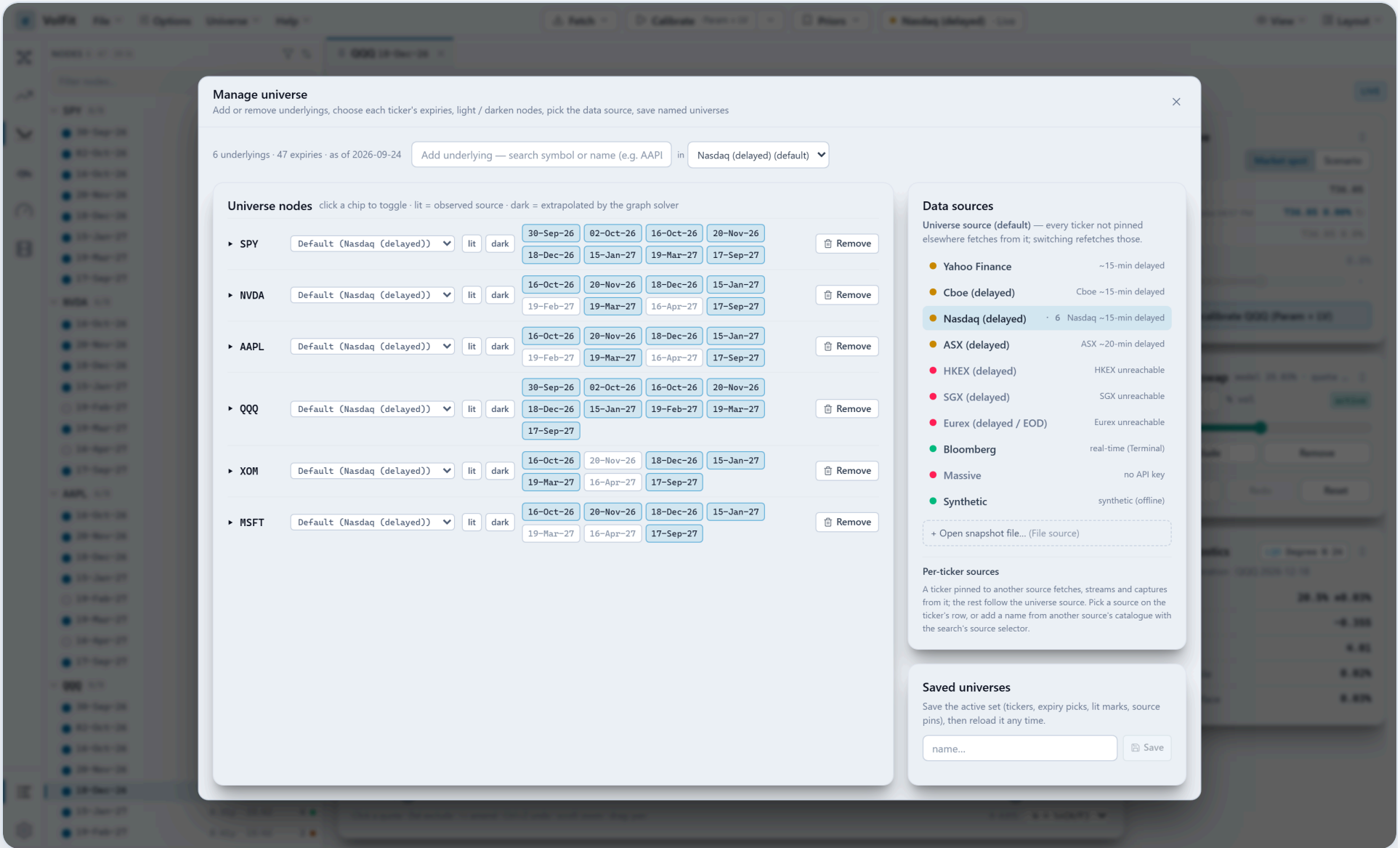
Universe, observations and fit state

The universe selection determines which smiles are observed and which are inferred.

- **Lit nodes:** Their current quotes enter calibration. Dark nodes receive an inferred result from the graph.
- **A calibration snapshot:** The fit retains the quotes, spot, forwards and settings used to produce it. Changed inputs can make it stale.
- **Separate operations:** Fetch refreshes observations. Calibrate re-estimates parameters. Spot updates transport the stored fit.

IN THE APP

Open Manage universe, identify one lit and one dark expiry, then show the Calibrate scope menu and the stale indicator.



24 September capture: six underlyings, 47 selected expiries. Filled and outlined expiry controls identify lit and dark nodes.

Snapshots, streams and vendor limits

A request pulls the whole chain at one instant; a stream keeps a local book current, contract by contract. Each vendor caps the live set and fills the rest in its own way.

	MASSIVE	BLOOMBERG
Request	REST snapshot: 250 contracts per page, 2 expiries in flight	Reference request (bdp): BID and ASK per contract
Request limit	None stated on the paid plan; 40 calls in flight ($\approx$ 110 per second) drew no 429	Metered: hits = contracts $\times$ fields; $\approx$ 4–5k unique securities a month on this account
Live channel	Websocket: every NBBO change, no conflation	Terminal subscriptions: conflated to 1 s (focus) or 5 s (the rest)
Live set limit	$\approx$ 1,000 contracts held per connection, one connection; app cap 950	App budget of 3,000 concurrent subscriptions
Over the cap	REST snapshot per ticker every 60 s (floor 15 s)	Rotation through 300 reserved slots, $\approx$ 15–18 contracts per second

MEASURED, 23–24 SEPTEMBER

Request: Massive returns 9 SPY expiries (3,454 quotes) in 1.2 s; Bloomberg returns 2 SPY expiries (900 contracts) in 5.8 s for 1,800 hits. Stream: a full SPY book reads into a chain in 42 ms; 420 NVDA contracts delivered 18,842 quotes in 8 s.

IN THE APP

Universe ▶ Manage universe (Ctrl+Shift+U), Data sources card: hover the Massive row for acknowledged / cap / msg/s / rest 60 s. The Quote Table badge reads LIVE or 1-min REST.

Data sources

Universe source (default) — every ticker not pinned elsewhere fetches from it; switching refetches those.

● Yahoo Finance

~15-min delayed

● Cboe (delayed)

· 1 Cboe ~15-min delayed

● Nasdaq (delayed)

Nasdaq ~15-min delayed

● ASX (delayed)

ASX ~20-min delayed

● HKEX (delayed)

HKEX unreachable

● SGX (delayed)

SGX unreachable

● Eurex (delayed / EOD)

Eurex unreachable

● Bloomberg

real-time (Terminal)

● Massive

· 1 delayed feed · streaming 420 · ...
420 acked · 98 msg/s · last tick 0 s

420 acked of 420 subscribed (cap 950)
98 msg/s · last tick 0 s · delayed cluster
NVDA 420/420 · focus none · rest 60 s

Per-ticker sources

1 pinned · [clear](#)

A ticker pinned to another source fetches, streams and captures from it; the rest follow the universe source. Pick a source on the ticker's row, or add a name from another source's catalogue with the search's source selector.

NVDA → Massive

Live desk, 24 September: NVDA on Massive, 420 of 420 contracts acknowledged under the 950 cap, 98 messages per second; the wings come from the 60 s REST layer.

Synchronising quotes to the current spot

An older quote was priced against the forward of its own moment. Invert it at that forward, then carry it to the current spot with the app's spot–vol rule.

THREE STEPS FOR A QUOTE i

1 · THE ERROR WE AVOID: INVERTING AT TODAY'S FORWARD

$$\varepsilon_i = \hat{\sigma}_i - \sigma_i \simeq -\frac{\partial V/\partial F}{\partial V/\partial \sigma} (F_{\text{now}} - F_i)$$

2 · INVERT AT F_i , THEN MOVE TO F_{now}

$$w_i^{\text{sync}} = w_i + \bar{w}(K_i | F_{\text{now}}) - \bar{w}(K_i | F_i)$$

THE REFERENCE SMILE, READ AT FORWARD F

$$\bar{w}(K | F) = w_{\text{ref}} \left(\log \frac{K}{F} + R \log \frac{F}{F_{\text{ref}}} \right)$$

3 · AGE ALLOWANCE

$$s_i = a \sqrt{t_{\text{now}} - t_i} \cdot \frac{\sigma_{\text{ATM}}}{15\%}$$

- **Step 2 in words:** each quote keeps its gap to the reference smile; only the reference moves, from F_i to F_{now} , by the rule R . Under sticky strike ($R = 1$) the correction is zero and only the label $\log(K/F)$ changes.
- **Step 3 in the fit:** a band fit widens the IV band by s_i on each side; a mid fit multiplies the quote's weight by $1/(1 + (s_i/\eta_i)^2)$, so a quote whose age allowance equals its half-spread counts half.

IN THE APP Options ▶ Calibration: the Quote synchronisation switch and the age scale a , in vol bp per $\sqrt{\text{minute}}$.

SYMBOL	MEANING
K_i, t_i, S_i	Strike of quote i , the time it was made, the spot at that time
$t_{\text{now}}, S_{\text{now}}$	Time and spot of the chain being fitted
F_i, F_{now}	Forwards at S_i and at S_{now} , same carry: $F_i = F_{\text{now}} S_i/S_{\text{now}}$
$V; \sigma_i, \hat{\sigma}_i$	Option price; its implied vol inverted at F_i (correct) and at F_{now} (naive)
w_i	Total variance $\sigma_i^2 \tau$ of the quote at F_i — bid, mid and ask alike
$w_{\text{ref}}, F_{\text{ref}}$	The node's last committed fit, a function of $\log(K/F_{\text{ref}})$, and its forward
R	Spot–vol rule: 0 sticky moneyness, 1 sticky strike, 2 sticky local vol
a, η_i	Age scale, 3.6 vol bp per $\sqrt{\text{minute}}$ at 15% ATM vol; the quote's half-spread in IV

WORKED NUMBERS

At the money the ratio in step 1 is $N(d_1)/(F\varphi(d_1)\sqrt{\tau}) \approx \sqrt{\pi/2}/(F\sqrt{\tau})$. Forward up 0.1% since the quote: $\varepsilon \approx -43$ vol bp at one month, -240 at one day. A 4-minute-old quote at 15% ATM: $s_i = 3.6\sqrt{4} = 7.2$ vol bp; with $\eta_i = 10$ bp its mid weight falls to $1/(1 + 0.72^2) \approx 0.66$.

Forward and discount from put–call parity

Every strike quoted on both sides gives one point on a line: its slope is the discount factor, its zero crossing the forward. On American chains the line is first cleaned of early-exercise premia.

PER EXPIRY, OVER THE n CALL–PUT PAIRS j

1 · REGRESSION, ONE VOTE PER PAIR

$$C_j - P_j = \alpha + \beta K_j + e_j, \quad \hat{D} = -\hat{\beta}, \quad \hat{F} = \hat{\alpha} / \hat{D}$$

2 · TRIM STALE PAIRS (AT MOST 3 ROUNDS, AT LEAST 3 PAIRS KEPT)

$$\text{drop } j \text{ if } |e_j| > 4 \max(1.4826 \operatorname{med}_k |e_k|, 10^{-4} S)$$

3 · AMERICAN CHAINS: RE-READ F FROM EUROPEAN PRICES, $\hat{D}$ HELD

$$F^{(m+1)} = \frac{1}{|A|} \sum_{j \in A} \left(K_j + \frac{C_j^E(F^{(m)}) - P_j^E(F^{(m)})}{\hat{D}} \right)$$

C_j, P_j : call and put mids at strike K_j ; S : spot; e_j : residual. A : the pairs nearest the money, at most 11 with $|\log(K_j/F)| \leq 0.15$. $C_j^E(F), P_j^E(F)$: European prices at the vols de-Americanised with the carry implied by F : $r = -\log(\hat{D})/t$, $q = r - \log(F/S)/t$.

IN THE APP

Forwards lens (Alt+2): the ladder's Parity, Theo and Active columns and the residual count per expiry. On the smile, puts and calls should join at the money.

QUESTION	IN THE CODE
Which strikes	Every strike where the call and the put both have a two-sided, uncrossed quote; at least 3. No moneyness window; quote edits and fit filters act later.
Weights	Equal: one vote per pair. The fit's quote-weighting scheme and the bid–ask widths are not used.
Absurd slope	$\hat{D}$ is kept to rates within $[-5\%, 30\%]$. At a bound, F is re-read from the level $K_j + (C_j - P_j)/D$, favouring tight, near-the-money pairs.
American: forward	Step 3 until F moves less than 0.5 bp: a median of 3 rounds on 81 real expiries, 6 at most.
American: rate	If call and put vols still differ by more than 5 vol bp at F , bisect r in $[-5\%, 30\%]$, rerunning step 3 each time: needed on 33 of the 81, median 4 steps (cap 24).
Then, per quote	One de-Americanisation of every quote at the final (F, D) : 24 bisections on a 192-step tree. No further loop.

RECORDED, SPY JUNE 2027 (0.93 YEARS)

Raw parity gave $\hat{D} = 1.0039$, a negative rate, and a 28 vol bp jump between puts and calls at the money. After step 3 and the rate search the jump is 1.1 vol bp. The whole de-bias of SPY's nine forwards took 182 ms.

Dividends, borrow and the joint carry fixed point

The theoretical forward is built from the rate and dividend inputs; the borrow is what makes it match the parity forward. On American chains the two depend on each other, so they are solved together.

PER EXPIRY, BORROW b SOLVED FROM $b_0 = 0$

1 · THEORETICAL FORWARD, FROM THE INPUTS

$$F_{\text{theo}}(b) = \left(S - \text{PV}_{\text{cash}}\right) e^{(r-q-b) t}, \qquad \text{PV}_{\text{cash}} = \sum_{t_k < t} d_k e^{-r t_k}$$

2 · PARITY FORWARD, AFTER DE-AMERICANISING AT CARRY $q + b$

$$C_j^E(b) - P_j^E(b) = D \left(F_{\text{par}}(b) - K_j\right) + e_j$$

3 · UPDATE UNTIL THE TWO FORWARDS AGREE

$$b_{n+1} = b_n + \frac{1}{t} \log \frac{F_{\text{theo}}(b_n)}{F_{\text{par}}(b_n)}$$

4 · WHAT THE READ IS WORTH

$$\sigma_b \approx \frac{\text{rms}(e)}{S t \sqrt{n}}, \qquad \left|\frac{\partial \sigma_{\text{ATM}}}{\partial b}\right| \approx \sqrt{\frac{\pi}{2}} \sqrt{t} \approx 1.25 \sqrt{t}$$

- **Why iterate:** the borrow changes the early-exercise premia, which move the parity forward. At the solution the de-Americanised call and put vols agree and parity holds at F_{theo} .
- **Stopping and use:** stop when b moves less than 0.1 bp a year: a median of 5 passes on 81 real expiries, 8 at most; a European chain needs one. The fit uses it only with Joint carry on and $|b| \geq 25$ bp.

SYMBOL	MEANING
S, t	Spot; calendar time to expiry in years (ACT/365)
r, q	Risk-free rate and continuous dividend yield, both inputs
d_k, t_k	Cash dividend amounts and their ex-dates before expiry, inputs
b	Borrow (stock-loan) cost, continuous: the unknown; positive means hard to borrow
$C_j^E(b), P_j^E(b)$	Call and put at strike K_j , de-Americanised on the tree at $(r, q + b)$ with the cash schedule, then repriced European
$F_{\text{par}}(b), D$	Forward and discount factor of that parity regression
$\text{rms}(e), n$	Residual size and number of pairs of the regression

RECORDED, XOM, 24 SEPTEMBER

Rate and dividends left at 0: $b = -331 \pm 18$ bp at 3 weeks, -124 ± 3 bp at 6 months. That is a carry mismatch, not a borrow: b absorbs the missing rate and dividends. The uncertainty shrinks like $1/t$.

Removing the early-exercise premium

Find the volatility at which the American tree reprices the quoted mid. At that volatility, the premium is the American price minus the European one.

PER QUOTE i , OUT-OF-THE-MONEY SIDE ONLY

1 · INVERT THE AMERICAN MID ON THE TREE

$$A_i^{\text{tree}}(\sigma_i^*) = A_i^{\text{mid}}$$

2 · THE PREMIUM, REMOVED FROM BID, MID AND ASK ALIKE

$$\widehat{\text{EEP}}_i = \max(A_i^{\text{mid}} - V_i^{\text{Black}}(\sigma_i^*), 0), \quad X_i^E = X_i - \widehat{\text{EEP}}_i$$

3 · CASH DIVIDENDS: THE ESCROWED SPOT AT TREE STEP m , NODE j

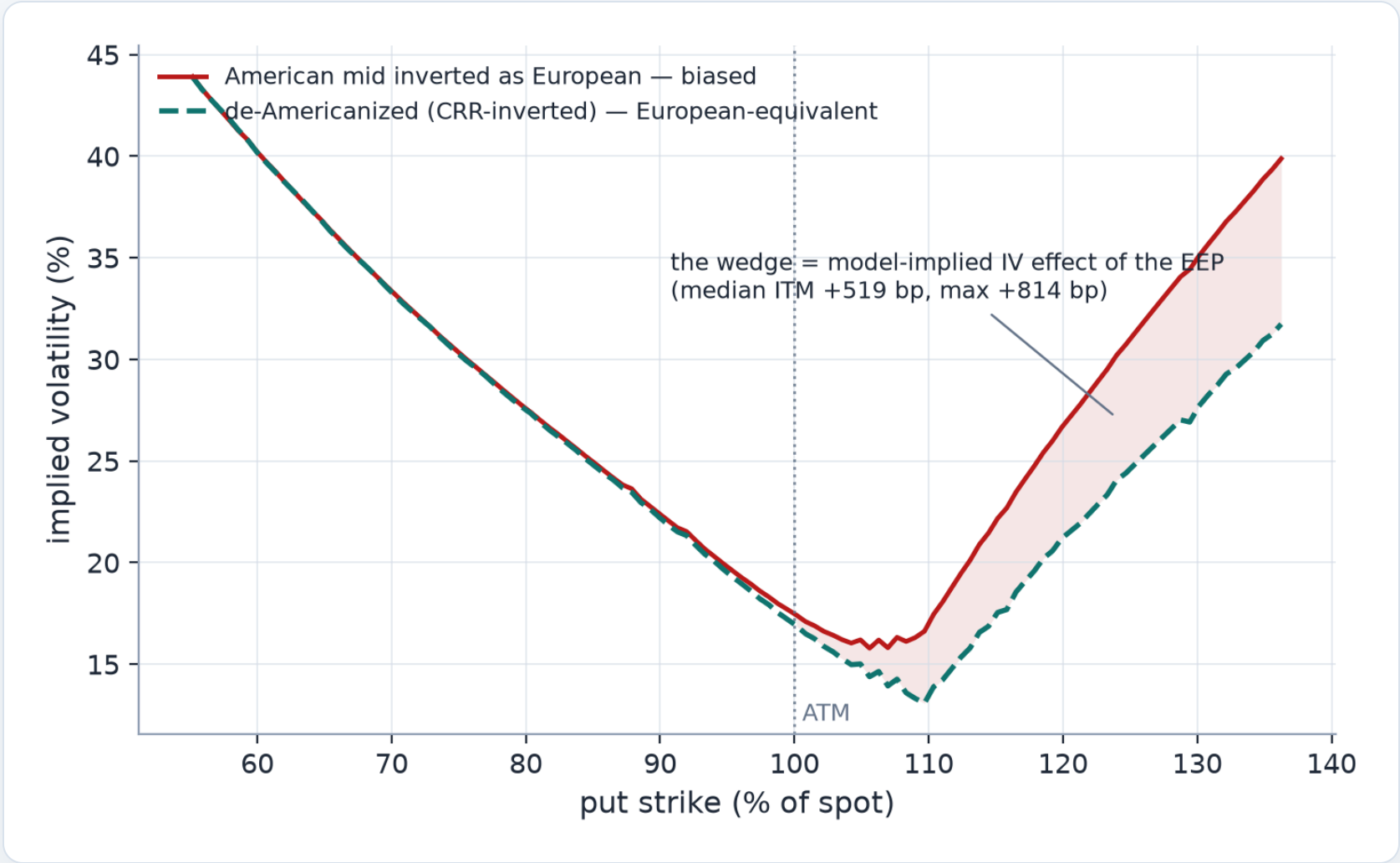
$$S_{m,j} = (S - \text{PV}(0)) u^{2j-m} + \text{PV}(t_m), \quad \text{PV}(s) = \sum_{s < t_k \leq t} d_k e^{-r(t_k-s)}$$

A_i^{mid} : American mid at strike K_i . $A_i^{\text{tree}}(\sigma)$: its price on the 192-step tree at volatility σ . $V_i^{\text{Black}}(\sigma)$: the European price at the resolved F and D . X_i : bid, mid or ask. $u = e^{\sigma\sqrt{t/N}}$, $t_m = mt/N$. d_k, t_k : cash dividends and ex-dates before expiry t .

- **The escrowed spot:** the stock is split into the dividends still to be paid, valued as riskless, and the rest. The tree diffuses the rest, so it recombines; exercise is tested against the full spot $S_{m,j}$.
- **Without a cash schedule:** the tree runs on the carry of the resolved forward:
 $r = -\log(D)/t$, $q = r - \log(F/S)/t$.

IN THE APP

Forward panel: switch the Dividend model to continuous with $q = 0$ and Apply — call and put markers split at the money; restore the schedule and Calibrate.



Recorded SPY chain, put strikes in % of spot: red, American mids inverted as if European; green, de-Americanised. The wide wedge is deep in-the-money puts, never fitted. On the fitted quotes the correction was a median 4.2 vol bp, at most 60.

Exercise correction: numerical cost and accuracy

The root is solved about a thousand times finer than the tree's own error, and a compiled, parallel kernel makes that affordable on every quote.

TREE WITH $N = 192$ STEPS, $\Delta t = t/N$

1 · UP FACTOR AND PROBABILITY

$$u = e^{\sigma \sqrt{\Delta t}}, \quad p = \frac{e^{(r-q)\Delta t} - u^{-1}}{u - u^{-1}}$$

2 · BACKWARD STEP: CONTINUE OR EXERCISE

$$V_{m,j} = \max\left(e^{-r\Delta t} \left[p V_{m+1,j+1} + (1-p) V_{m+1,j}\right], \Pi(S_{m,j})\right)$$

3 · ROOT BY BISECTION (DICHOTOMY), 24 HALVINGS

$$\sigma_{\text{mid}} = \frac{1}{2}(\sigma_{\text{lo}} + \sigma_{\text{hi}}) : A^{\text{tree}}(\sigma_{\text{mid}}) < A^{\text{mid}} \Rightarrow \sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}, \text{ else } \sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$$

4 · DISCRETISATION CHECK: A ROUND TRIP FROM A NEAR-EXACT PRICE

$$\varepsilon_{\text{tree}}(K) = \sigma_{192}^* \left(A_{2001}^{\text{tree}}(\sigma_0; K) \right) - \sigma_0$$

$V_{m,j}$: option value at step m after j up-moves. Π : exercise payoff, $(S - K)^+$ or $(K - S)^+$. $S_{m,j}$: the escrowed spot of the previous slide. Bracket: $\sigma_{\text{lo}} = \max(10^{-4}, 1.5|r - q|\sqrt{\Delta t})$, $\sigma_{\text{hi}} = 0.5$ doubled up to 4 until it clears the quote; after 24 halvings $\sigma_{\text{hi}} - \sigma_{\text{lo}} \leq 4 \cdot 2^{-24}$, 0.002 vol bp. $A_N^{\text{tree}}(\sigma_0; K)$: American price on an N -step tree at the known σ_0 ; $\sigma_{192}^*(\cdot)$: the production inversion.

- **When it compiles:** on the first call in each process (Numba, just in time): about 1.5 s from scratch. The machine code is cached on disk; a restart or a pool worker loads it in about 0.2 s.
- **What forces a recompile:** a change of the kernel's source, of Numba or Python, or of the CPU. Tree depth, bisection count, strikes and chain size are values, not types: changing them never recompiles.

STEP	HOW OFTEN	COST
Screen, escrow $PV(t_m)$, $e^{(r-q)\Delta t}$, $e^{-r\Delta t}$, σ_{lo}	Once per expiry	Shared by all its quotes
Scratch arrays for $V_{m,\cdot}$ and u^k	Once per quote	Reused by its 26 tree prices: floor, bracket, 24 halvings
u , p and the powers u^{k-N}	Once per tree price	Two exps, $2N$ multiplies
Backward step	18,721 nodes per tree	Multiply, add, max: no exp
Quotes	In parallel	20 threads on this machine
Prepared quotes	Per new chain, forward, carry or clock	Cached: a fit-setting change reuses them

MEASURED, 29 SEPTEMBER

Round trip at $\sigma_0 = 25\%$ ($S = 100$, $r = 4.5\%$, $q = 1.2\%$, $t = 0.5$, strikes 70–130): within ± 5.4 vol bp at $N = 192$ (the 70 put +5.4, ATM +3.6, most under 2), within ± 2.1 at $N = 501$. Speed: 300 quotes in 17 ms, 0.056 ms each; the NumPy version of the same algorithm takes 1.29 s, same results.

The units of a calibration residual

Every model is fitted by least squares on per-quote residuals. Dividing each price error by the quote's vega makes every residual read in vol units, at every strike.

PER QUOTE i , MODEL PARAMETERS θ

1 · RESIDUAL: MODEL PRICE ERROR OVER VEGA, FLOORED

$$r_i(\theta) = \frac{c_\theta(k_i) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta}$$

2 · WHAT IT READS: THE VOL MISS, PLUS A VOMMA TERM FAR FROM THE FIT

$$r_i = \Delta\sigma_i \left(1 + \frac{1}{2} \frac{d_+ d_-}{\sigma_i} \Delta\sigma_i + O(\Delta\sigma_i^2) \right), \quad \Delta\sigma_i = \sigma_\theta(k_i) - \sigma_i$$

3 · OBJECTIVE (MID TARGET; BANDS ON THE NEXT SLIDE)

$$\min_{\theta} \sum_i \lambda_i r_i(\theta)^2 + \text{model penalties}$$

- **Price-space models:** LQD and local volatility produce c_θ as a price, which keeps every iterate a valid density; the division by vega moves the error to vol units.
- **Vol-space models:** SVI and MCS produce σ_θ directly: their residual is exactly $\Delta\sigma_i$.
- **The floor η :** where vega is tiny — far wings, last days — the residual stops amplifying and behaves like a scaled price error.

SYMBOL	MEANING
k_i	Log-moneyness $\log(K_i/F)$ of quote i
σ_i, w_i	Market implied vol and total variance $w_i = \sigma_i^2 \tau$
$B(k, w)$	Black call price in forward units: price / (DF)
$c_\theta(k), \sigma_\theta(k)$	Model call price in the same units, and its implied vol
$\partial_\sigma B$	Black vega at the market vol, $\varphi(d_+) \sqrt{\tau}$; fixed per quote during the fit
$d_\pm$	$-k/\sqrt{w} \pm \sqrt{w}/2$
η	Vega floor, 10^{-4}
λ_i	Quote weight: one vote per quote by default

ILLUSTRATIVE CALCULATION

A 0.04 USD price error on an option with a vega of 20 USD per unit of volatility (0.20 USD per vol point): $r = 0.04/20 = 0.002$, that is 20 vol bp. The currency cancels. Far from the fit the vomma term appears: a 100 vol bp miss at $k = 0.2$ ($\sigma = 25\%$, $\tau = 0.25$) reads as 105 vol bp; at the money it reads as 100.

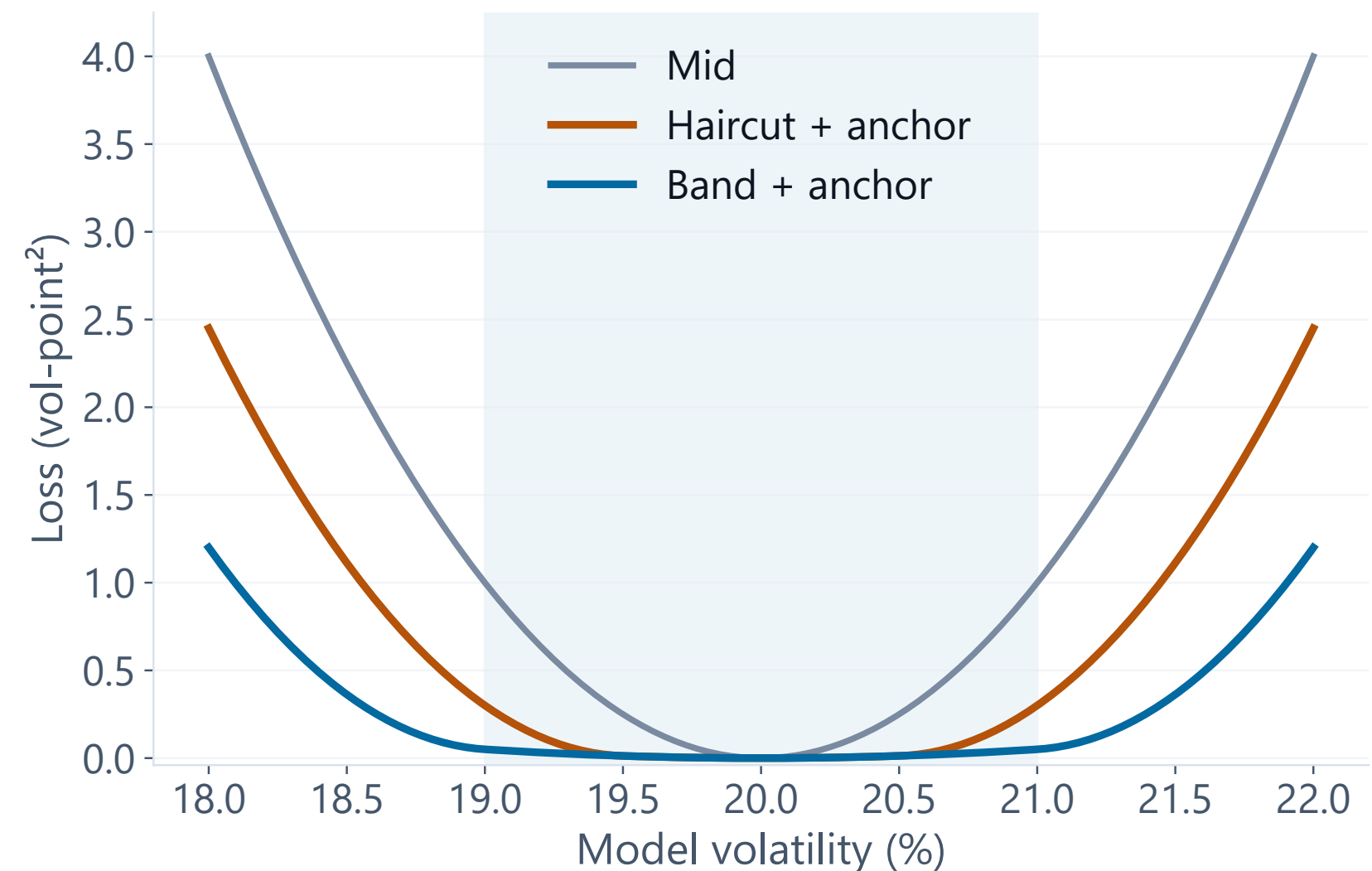
Mid, bid–ask and haircut targets

A band target allows a range of fitted values, with a weak mid anchor selecting among them.

$$L_i = (m_i - u_i)_+^2 + (l_i - m_i)_+^2 + \theta_{\text{mid}}(m_i - \bar{m}_i)^2$$

- **Mid:** Minimise the squared difference to the quoted mid.
- **Band:** Penalise values outside the interval. Inside it, only the mid-anchor term remains.
- **Haircut:** Move each side toward mid by h , stopping at mid. A spread narrower than $2h$ collapses to a mid target.

IN THE APP Show the target shading on the Smile, then the Mid / Bid–Ask / Haircut controls in Calibration options.



Illustrative 19% / 21% quote. A 0.5 vol-point haircut leaves [19.5%, 20.5%]. The band includes a 0.05 mid-anchor weight.

Quote weights and strike spacing

The weights decide which parts of the smile the fit cares about: the exchange's listing (equal weights), or a chosen target.

WEIGHT λ_i OF QUOTE i AT LOG-MONEYNESS k_i

1 · TARGET SCHEMES: TARGET SHAPE $\times$ SPACING, CAPPED, MEAN ONE

$$\lambda_i \propto f(k_i) \min\left(\frac{s_i}{\bar{s}}, 10\right), \quad \frac{1}{n} \sum_i \lambda_i = 1$$

2 · SPACING: WIDTH OF THE QUOTE'S VORONOI CELL

$$s_i = \frac{1}{2} (k_{i+1} - k_{i-1})$$

3 · EQUAL WEIGHTING, THE DEFAULT: ONE VOTE PER QUOTE

$$\lambda_i = 1$$

- **Target f :** the shape the summed weight should follow along the smile: flat, option time value, vega or OTM delta.
- **Voronoi cell:** the stretch of log-moneyness closer to this quote than to any other; its width s_i is half the gap to each neighbour, so dense clusters get narrow cells.
- **Equal weighting:** every quote counts once, with no spacing correction. The summed weight then follows the exchange's listing — heaviest where strikes are densest, usually near the money.

ILLUSTRATIVE CALCULATION

Near the money, three quotes 0.01 apart; in the wing, one quote alone in a stretch of 0.03. Equal weights give the cluster 3 votes to 1. With the flat target, each clustered quote stands for 0.01 and the lone quote for 0.03, so both stretches of the smile weigh the same.

IN THE APP

Smile view: switch the Weights layer on in the layer rail — the strip under the chart shows the target beside the weight the fit sums; hover a bar for $k \cdot \text{target} \cdot \times \text{spacing} \cdot \text{weight}$.



Reference-note chain: grey, equal weights, 1 per quote; green, the time-value target with the spacing correction. Isolated strikes gain weight, clustered ones share it, and the target fades the far wings.

What each model parameterises

The four families place flexibility in different mathematical objects.

MODEL	UNKNOWN OBJECT	PRICING / INVERSION	MAIN CONSTRAINTS OR CONTROLS
LQD	Log quantile density of log-return	Integrate a probability law	Moment integrability; numerical pricing checks
SVI-JW	Total implied variance $w(k)$	Black formula	Structural wing bounds; density checks
MCS	Variance profile with local corrections	Black formula	Wing budget; positivity and density checks
Local volatility	Local variance $\nu(\tau, x)$	Forward Dupire PDE	Positive vertices; roughness; lattice checks

COMPARISON CONVENTION

Use the same prepared quotes, forward, variance clock, target mode and weighting. Record the tail and regularisation choices alongside the fit error.

LQD: constructing a probability law

LQD models the distribution of the log-return through the slope of its quantile. Any real function g gives a valid law: one shift and one inequality do the rest.

LOG-RETURN $X = \log(S_T/F_T)$, QUANTILE $Q(u)$ AT RANK $u \in (0, 1)$

1 · QUANTILE SLOPE: POSITIVE FOR EVERY g , SO THE DENSITY IS TOO

$$Q'(u) = q(u) = \frac{e^{g(u)}}{u(1-u)} > 0, \quad f_X(Q(u)) = \frac{1}{q(u)}$$

2 · THE SMOOTH PART g : ENDPOINT TERMS AND A LEGENDRE BODY

$$g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u)$$

3 · ONE SHIFT MAKES THE FORWARD THE MEAN

$$Q(u) = \mu + \int_{1/2}^u q(v) dv, \quad \mathbb{E}[e^X] = \int_0^1 e^{Q(u)} du = 1$$

4 · TAIL SCALES: THE QUANTILE'S SLOPE IN LOG-ODDS AT EACH END

$$A_L = e^{g(0)}, \quad A_R = e^{g(1)} : \quad \mathbb{P}(X < x) \asymp e^{x/A_L}, \quad \mathbb{P}(X > x) \asymp e^{-x/A_R}$$

- **Why $A_R < 1$:** the forward is the mean $\mathbb{E}[e^X]$. In the right tail the density falls like e^{-x/A_R} , so $e^x e^{-x/A_R}$ is integrable only if $A_R < 1$. The left tail needs no condition.
- **In the fit:** the solver moves $\log A_L$, ρ with $A_R = 1/(1 + e^{-\rho})$, and body modes that leave both tails fixed, so every point it visits is an admissible law.

IN THE APP

Parametric ▶ Density ▶ Log Q-density: the two logarithmic walls at the ends and the Legendre body between them; Density and CDF are one click away.

SYMBOL	MEANING
X, S_T, F_T	Log-return to expiry: terminal spot over the forward
$u, Q(u)$	Rank in $(0, 1)$ and the quantile: $\mathbb{P}(X \leq Q(u)) = u$
$q(u), f_X$	Quantile slope, and the density of X : at matching ranks $f_X = 1/q$
$g(u)$	Any real function: the model's parameters live here
L, R	Endpoint terms of g ; alone they are not the tail scales
a_n, P_n	Body coefficients on Legendre polynomials, $n = 2 \dots N$; $N = 16$ by default
μ	Shift fixing $\mathbb{E}[e^X] = 1$
A_L, A_R	Left and right tail scales: $A_L = e^{L + \sum_n a_n}$, $A_R = e^{R + \sum_n (-1)^n a_n}$

WORKED EXAMPLE

All $a_n = 0$ and $L = R = \log s$: then $g \equiv \log s$ and $X = \mu + s \log \frac{U}{1-U}$ with U uniform, a logistic law with $A_L = A_R = s$. An SPX-like fit has $A_L = 0.157$, $A_R = 0.038$: each further 0.157 of log-return down divides the probability by e , each 0.038 up does the same. The heavier left tail is the skew.

LQD: from the quantile to option prices

A call is exercised on every rank above its strike's rank. Its price is the stock those outcomes deliver minus the strike they pay: two running sums along the quantile.

STRIKE RANK u_k : $Q(u_k) = k$, SO $\mathbb{P}(X \leq k) = u_k$

1 · THE PAYOFF, RANK BY RANK: EXERCISED ON THE RANKS ABOVE u_k

$$c(k) = \mathbb{E}[(e^X - e^k)^+] = \int_{u_k}^1 (e^{Q(u)} - e^k) du$$

2 · SPLIT: ASSET SHARE MINUS STRIKE CASH

$$c(k) = G(u_k) - e^k (1 - u_k)$$

3 · THE ASSET SHARE: THE STOCK DELIVERED ABOVE RANK u

$$G(u) = \int_u^1 e^{Q(v)} dv = \mathbb{E}[e^X \mathbf{1}\{X > Q(u)\}], \quad G(0) = 1$$

4 · IF X IS NORMAL WITH VARIANCE w : THE TWO LEGS ARE BLACK'S

$$G(u_k) = \Phi(d_+), \quad 1 - u_k = \Phi(d_-) \implies c = B(k, w)$$

- **Asset share** $G(u)$: the part of the forward's value delivered by the outcomes above rank u . It falls from 1 at $u = 0$ to 0 at $u = 1$: an asset-or-nothing digital, in forward units.
- **Strike cash**: on exercise the holder pays the strike, $e^k = K/F$ per unit of forward, in the fraction $1 - u_k$ of outcomes: the strike times a cash-or-nothing digital. The price is what is received minus what is paid.
- **Every strike from one build**: Q and G are computed once per parameter set; each strike then costs one root, $Q(u_k) = k$, and two look-ups.

SYMBOL	MEANING
$c(k)$	Call price in forward units, $C/(DF)$, at $k = \log(K/F)$
$X, Q(u)$	Log-return $\log(S_T/F_T)$ and its quantile at rank u
u_k	The strike's rank: outcomes below it expire worthless
$1 - u_k$	Probability that the call finishes in the money
$e^{Q(u)}$	S_T/F_T in the outcome of rank u
$G(u)$	Asset share: the forward's value delivered above rank u
e^k	Strike in forward units, K/F
$\Phi(d_{\pm})$	Black's two legs, $d_{\pm} = -k/\sqrt{w} \pm \sqrt{w}/2$; $B = \Phi(d_+) - e^k \Phi(d_-)$

WORKED EXAMPLE

Logistic law, $A_L = A_R = 0.0813$: ATM vol exactly 20% at $\tau = 0.5$. Strike 110.52 on a forward of 100 ($k = 0.10$): rank $u_k = 0.7965$, so exercised in 20.35% of outcomes. Asset share 24.72, strike cash $110.52 \times 0.2035 = 22.49$: price 2.23, implied vol 20.60%.

LQD calibration: one build per step, exact derivatives

At each step the solver needs every quote's price and its derivative in every parameter. Both come from one build of the slice: running sums on a fixed grid, read at each strike. The strike's rank moves with the parameters, but that move adds nothing.

PARAMETERS $\theta = (L, R, a_2, \dots, a_N)$; BASIS $\varphi_j = \partial g / \partial \theta_j$

1 · A PRICE'S DERIVATIVE: THE MOVING LOWER LIMIT CONTRIBUTES NOTHING

$$\frac{\partial c(k)}{\partial \theta_j} = \int_{u_k}^1 e^{Q(u)} \frac{\partial Q(u)}{\partial \theta_j} du - \underbrace{(e^{Q(u_k)} - e^k)}_{=0} \frac{\partial u_k}{\partial \theta_j}$$

2 · THE QUANTILE'S DERIVATIVE: ONE MORE RUNNING SUM PER PARAMETER

$$\frac{\partial Q(u)}{\partial \theta_j} = \frac{\partial \mu}{\partial \theta_j} + D_j(u), \quad D_j(u) = \int_{1/2}^u q(v) \varphi_j(v) dv$$

3 · THE SHIFT'S DERIVATIVE: THE FORWARD STAYS THE MEAN

$$\frac{\partial \mu}{\partial \theta_j} = - \int_0^1 e^{Q(u)} D_j(u) du$$

φ_j is $1 - u$ for L , u for R and $P_n(1 - 2u)$ for a_n ; fixed functions, because g is linear in θ , so $\partial q / \partial \theta_j = q \varphi_j$. Line 3 differentiates $\int_0^1 e^{Q(u)} du = 1$. Line 1 is $\partial G / \partial \theta_j$ read at the fixed rank u_k .

- **Why it matters:** finite differences rebuild the slice once more per parameter and re-solve every strike's rank; the exact derivatives reuse the one build, with two more running sums per parameter.
- **Capacity:** the order is cut to two quotes per parameter, $N + 1 \leq n/2$ for n quotes, never below 6 (24 quotes: $N = 11$); a ridge $10^{-6} n^2 a_n^2$ on the modes $n \geq 4$ damps oscillation.

STAGE	WHAT THE CODE DOES
Start	Logistic law: $a_n = 0$, $L = R = \log s$, $s = \sqrt{3w_0}/\pi$ from the quotes' ATM variance w_0 ; in a surface sweep, the previous expiry's fit
Build, each trial θ	2,001 nodes even in log-odds $\log \frac{u}{1-u}$, from -40 to 40 , where the quantile's slope is e^g : running sums give Q , then μ , then G
Price, each quote	Newton on the interpolated Q for the rank u_k , then $G(u_k) - e^k(1 - u_k)$
Derivatives	Lines 1–3 for all $N + 1$ parameters, on the same grid, read at the same ranks
Solve	Trust-region least squares on the quote residuals, the ridge and the A_R barrier; stops at relative changes of 10^{-10}
Accept	One rebuild on 8,001 nodes for the prices, the density and the variance swap

MEASURED, 23 AND 29 SEPTEMBER

SPY 16 October expiry, 121 quotes (23 September): the default order-16 fit takes 26 ms, 14 evaluations, 2.06 vol bp RMS. On a 24-quote synthetic strip today ($N = 11$): 9 ms with the exact derivatives, 23 ms with finite differences, same optimum; the two Jacobians agree to 4×10^{-5} .

Two ways to summarise a smile: ATM handles and delta packages

Three derivatives at the money, or three option packages at fixed deltas. They agree to first order but are different coordinates: the filter and the graph carry the first, the prior's default persists the second.

IMPLIED VOL $\sigma(k)$ OF ONE EXPIRY; DELTA- d LEGS k_c (CALL) AND k_p (PUT)

1 · ATM HANDLES: LEVEL, SKEW, CURVATURE — EXACT ON LQD

$$(\sigma_0, s_0, \kappa_0) = (\sigma(0), \sigma'(0), \sigma''(0))$$

2 · DELTA PACKAGES: RISK REVERSAL AND BUTTERFLY

$$RR_d = \sigma(k_c) - \sigma(k_p), \quad BF_d = \frac{1}{2} [\sigma(k_c) + \sigma(k_p)] - \sigma_0$$

3 · THE LINK: A TAYLOR EXPANSION AT THE MONEY, EXACT ONLY FOR A QUADRATIC SMILE

$$RR_d \approx s_0 (k_c - k_p) + \frac{1}{2} \kappa_0 (k_c^2 - k_p^2)$$
$$BF_d \approx \frac{1}{2} s_0 (k_c + k_p) + \frac{1}{4} \kappa_0 (k_c^2 + k_p^2)$$

k_c solves $k = \frac{1}{2} \sigma(k)^2 \tau - \sigma(k) \sqrt{\tau} \Phi^{-1}(d)$: the strike whose forward Black call delta is d ($d = 0.25$ for 25Δ); k_p uses $1 - d$. The legs sit near $\pm 0.67 \sigma \sqrt{\tau}$ at 25Δ and $\pm 1.28 \sigma \sqrt{\tau}$ at 10Δ . Units: s_0 in vol per unit of k , κ_0 per unit of k^2 ; RR and BF in vol.

- **Not interchangeable:** the legs scale with $\sigma \sqrt{\tau}$ and move with the smile, and past second order the wings enter. The same three handles give different packages at different maturities, and RR or BF can move while the handles stay fixed.
- **Division of labour:** persistence fills gaps: its legs sit where quotes thin out, and it ignores a level move. The filter and the graph track what every chain measures, the money, with exact handles for their covariances and for rebuilding a smile.

IN THE APP

Options ▶ Prior: the Hybrid, Quote operators and Smile factors modes, the Operators chips and the Collar sign; Options ▶ Kalman filter; the Fit diagnostics card shows ATM, skew and curvature.

CONSUMER	CARRIES	WHY THIS SET
Kalman filter	σ_0, s_0, κ_0 of the LQD fit	Measured on every chain; exact, so the fit's information gives its noise; a spot move shifts it in closed form
Graph	σ_0, s_0, κ_0 , as three fields	Same carrier as the filter; the posterior is written back into an LQD smile, the node's other modes held
Prior, Hybrid (default)	ATM, RR25, BF25, var swap; tail anchors at 2, 5, 10 Δ	Acts only where unquoted; legs reach the wings; RR and BF weights sum to zero, so a level move passes through
Prior, Smile factors	ATM, skew, curvature from σ at 0, ± 0.06	Available; on liquid chains these legs are quoted, so the gates close

WORKED NUMBERS

SPX-like smile: $\sigma_0 = 21.8\%$, $s_0 = -0.178$, $\kappa_0 = 0.024$. At $\tau = 0.75$ the 25Δ legs sit at $k = +0.127$ and -0.118 : $RR_{25} = -4.28$ vol points, line 3 gives -4.34 . The same curve at one month: legs at ± 0.042 , $RR_{25} = -1.50$.

SVI and its Jump-Wings reading

Raw SVI draws total variance as a tilted hyperbola with two straight wings; Jump-Wings reads the same curve as five desk quantities.

TOTAL VARIANCE $w(k) = \sigma(k)^2\tau$ AT LOG-MONEYNESS k

1 · RAW SVI: A V WITH A ROUNDED BOTTOM, SHIFTED AND TILTED

$$w(k) = a + b \left[\rho(k - m) + \sqrt{(k - m)^2 + \xi^2} \right], \quad b > 0, \quad |\rho| < 1, \quad \xi > 0$$

2 · ITS WING SLOPES AND ITS LOWEST POINT

$$\beta_L = b(1 - \rho), \quad \beta_R = b(1 + \rho), \quad w_* = \min_k w = a + b\xi\sqrt{1 - \rho^2}$$

3 · JUMP-WINGS: FIVE READINGS OF THE SAME CURVE

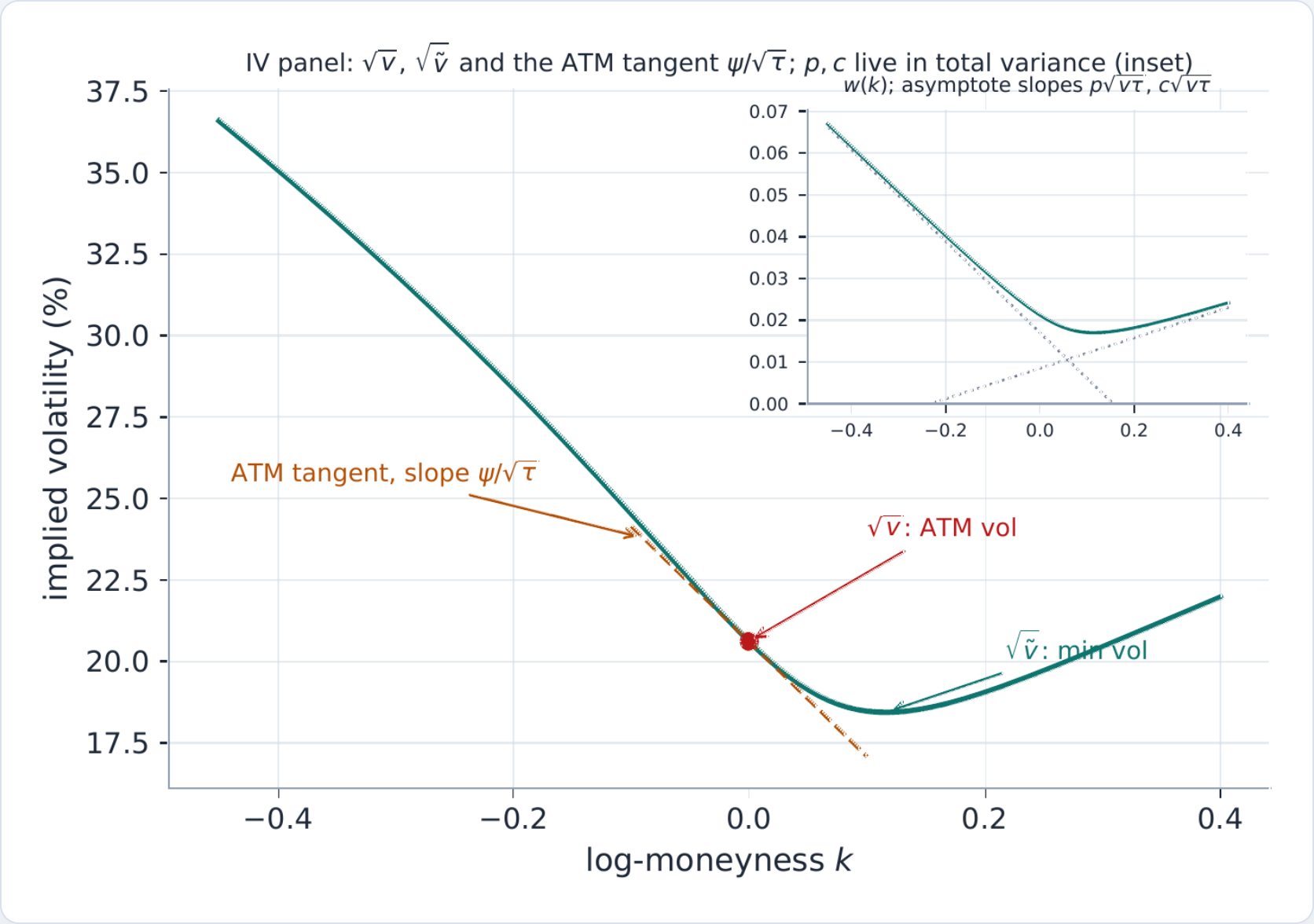
$$v = \frac{w_0}{\tau}, \quad \psi = \frac{w'(0)}{2\sqrt{w_0}}, \quad p = \frac{\beta_L}{\sqrt{w_0}}, \quad c = \frac{\beta_R}{\sqrt{w_0}}, \quad \tilde{v} = \frac{w_*}{\tau}$$

$w_0 = w(0)$. a lifts the curve, b scales both wings, ρ tilts them, m is the centre of the square root (not the minimum), ξ rounds the bottom. $\sqrt{v}$: ATM vol; $\psi/\sqrt{\tau} = \sigma'(0)$, the ATM skew s_0 ; p, c : wing slopes over $\sqrt{w_0}$ (JW's names; c is not a price here); $\sqrt{\tilde{v}}$: lowest vol.

- **Fitting chart:** raw SVI is stored; the solver moves β_L, β_R , the bottom (k_*, w_*) and its curvature κ_* , keeping both wings below 1.95 (Lee's limit is 2) and $w_* > 0$.
- **Convex is not arbitrage-free:** $w'' > 0$, but the density needs w, w', w'' together: checked on 801 points across the quotes; a failing fit is refitted once.

WORKED EXAMPLE

SPX-like smile, $\tau = 0.75$: $a = 0.0096, b = 0.0427, \rho = -0.68, m = 0.283, \xi = 0.306$. Read in JW: $\sqrt{v} = 21.8\%, \psi/\sqrt{\tau} = -0.178, p = 0.380, c = 0.072, \sqrt{\tilde{v}} = 16.0\%$.



Reference-note curve read in Jump-Wings: ATM vol $\sqrt{v}$, ATM tangent of slope $\psi/\sqrt{\tau}$ and lowest vol $\sqrt{\tilde{v}}$ on the vol chart; the wing slopes $p\sqrt{v\tau}$ and $c\sqrt{v\tau}$ only in total variance (inset).

MCS: a convex base plus local corrections

A convex base owns the wings and the overall skew; up to two signed hats add a shoulder or a notch in the body and vanish in both wings. Not arbitrage-free by construction: the core count, a ridge, a wing penalty and the density check govern it.

IMPLIED VARIANCE $v = \sigma^2$ AGAINST $z = k/(\sigma_{\text{ref}}\sqrt{\tau})$

1 · LOG-COSH: A ROUNDED ABSOLUTE VALUE, STRAIGHT FAR OUT

$$\Phi_{\gamma}(x) = \frac{4}{\gamma^2} \log \cosh \frac{\gamma x}{2} \approx \frac{1}{2}x^2 \text{ near } 0, \quad \Phi'_{\gamma}(x) \rightarrow \pm \frac{2}{\gamma}$$

2 · MODEL: BASE (VALUE, SLOPE, CURVATURE AT z_0) PLUS $n_c \leq 2$ SIGNED HATS

$$v(z) = \underbrace{v_0 + v_1(z - z_0) + v_2 \Phi_{\gamma_{\pm}}(z - z_0)}_{\text{base}} + \sum_{j=1}^{n_c} \alpha_j B_j(z)$$

3 · HAT: A SECOND DIFFERENCE OF THE SAME LOG-COSH, HEIGHT 1 AT m_j

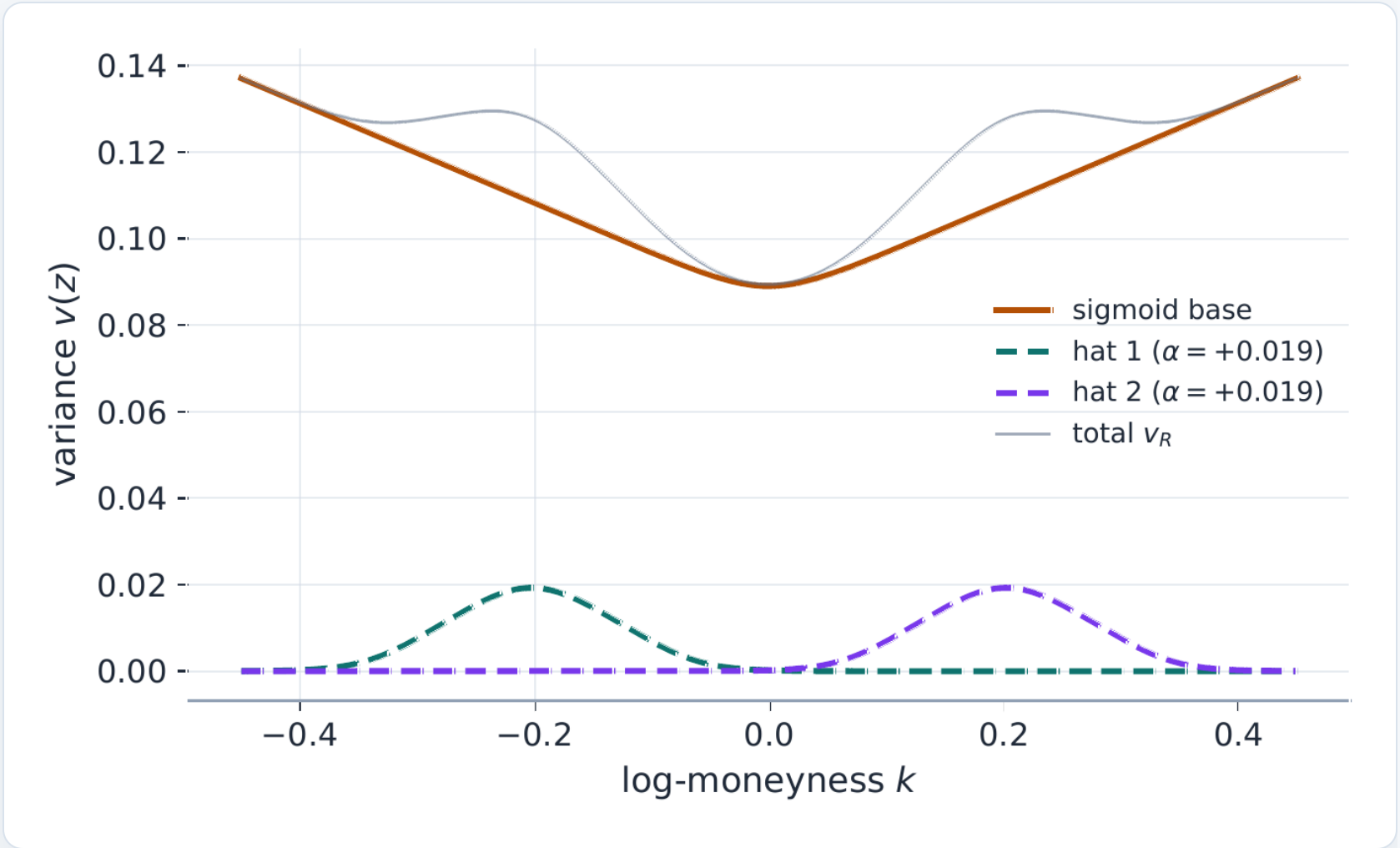
$$B_j(z) = \frac{\Phi_{\gamma_j}(z - m_j - h_j) - 2 \Phi_{\gamma_j}(z - m_j) + \Phi_{\gamma_j}(z - m_j + h_j)}{2 \Phi_{\gamma_j}(h_j)}$$

σ_{ref} : quoted vol nearest the money; $w = \tau v$. γ_{-}, γ_{+} : how fast the left and right wings straighten; z_0 is fitted, so v_0, v_1 are not the ATM level and skew. Hat j : height α_j , centre m_j , half-width h_j , sharpness γ_j .

- Wings untouched: a second difference of a straight line is zero, and Φ is straight far out: each hat, its slope and its curvature vanish in both wings. The wing slopes stay the base's, $v_1 \mp 2v_2/\gamma_{\mp}$.

ILLUSTRATIVE CALCULATION

A hat of height $\alpha = 0.019$ on a 30% vol lifts its centre to $\sqrt{0.09 + 0.019} = 33.0\%$. Seed shape $h = 0.4$, $\gamma = 5$: 4% of it is left one unit of z away, 0.03% at two.



Reference-note decomposition: the convex base (orange) and two hats of height +0.019 (dashed) add up to the W-shaped total (grey, v_R in the figure); far out the total rejoins the base.

Generalised LQD tail exponents

One exponent per side moves each LQD tail from exponential to Gaussian decay. The quoted strikes barely see it; the moments and the far wings do.

RANK u , LOG-ODDS $z = \log \frac{u}{1-u}$, $Y = S_T/F_T$, LOG-RETURN $X = \log Y$

1 · QUANTILE SPEED: SLIDE 14'S e^g TIMES ONE GAUGE PER SIDE

$$\frac{dQ}{dz} = e^{g(u)} (1 - \log u)^{-\alpha_-} (1 - \log(1 - u))^{-\alpha_+}, \quad 0 \leq \alpha_{\pm} \leq \frac{1}{2}$$

2 · RIGHT TAIL OF X , TO LEADING ORDER AS $x \rightarrow \infty$ (THE LEFT MIRRORS IT WITH A_L , α_-)

$$\log \mathbb{P}(X > x) \sim -\left(\frac{(1 - \alpha_+)x}{A_R}\right)^{\frac{1}{1-\alpha_+}}$$

3 · FAR RIGHT WING OF TOTAL VARIANCE, FOR $\alpha_+ > 0$

$$w(k) \sim \frac{1}{2} \left(\frac{A_R}{1 - \alpha_+}\right)^{\frac{1}{1-\alpha_+}} k^{\frac{1-2\alpha_+}{1-\alpha_+}}$$

g , $A_L = e^{g(0)}$, $A_R = e^{g(1)}$; as on slide 14. The gauge $1 - \log u$ grows like $1 + |z|$ in the left tail and tends to 1 in the right; $1 - \log(1 - u)$ does the reverse, so each exponent acts on its own tail. $w(k)$: total implied variance at log-moneyness k . $\sim$: the ratio tends to one.

- **Policy:** one pair (α_-, α_+) per underlying, shared by all its expiries and fixed before the fit. The default $\alpha_{\pm} = 0$ reproduces the exponential model bit for bit.
- **Why α is not fitted:** a small $\alpha > 0$ moves the quoted smile by a few bp but makes every moment finite. The quotes cannot choose the class: compare scenarios instead.

α · PRESET	LOG-RETURN TAIL (RIGHT SIDE)	FAR-WING TOTAL VARIANCE
0 · Exp	Exponential, e^{-x/A_R} ; $\mathbb{E}[Y^r]$ finite only for $r < 1/A_R$	Linear in k , at the Lee slope of the next slide
0.25 · Int	$e^{-cx^{4/3}}$ with $c = (0.75/A_R)^{4/3}$; every $\mathbb{E}[Y^r]$, $r > 0$, finite	Sublinear, $\propto k^{2/3}$; Lee slope 0
0.5 · Gauss	Gaussian rate, $e^{-x^2/(4A_R^2)}$; a normal tail of variance $2A_R^2$	Tends to the constant $2A_R^2$; Lee slope 0

RECORDED SCENARIO RUN · 13 AUGUST 2026

AAPL 18-Sep-26 (0.17 y), same quotes, refitted with $\alpha_+ = 0$ and with $\alpha_{\pm} = 1/2$: strip RMS 12.6 vs 12.5 vol bp, var-swap vol 32.06% vs 32.01%. The moments $\mathbb{E}[Y^r]$ are finite only for $-3.95 < r < 19.0$ at $\alpha = 0$, for every r at $\alpha = 1/2$.

Wing slopes and finite moments

How steeply total variance grows in the far wings is set by how many moments the terminal price has. Lee's formula makes the link exact, for any model.

$Y = S_T/F_T$ WITH $\mathbb{E}[Y] = 1$; TOTAL VARIANCE $w(k)$ AT $k = \log(K/F)$

1 · WING SLOPES: GROWTH OF TOTAL VARIANCE PER UNIT OF $|k|$, FAR OUT

$$\beta_R = \limsup_{k \rightarrow +\infty} \frac{w(k)}{k}, \quad \beta_L = \limsup_{k \rightarrow -\infty} \frac{w(k)}{|k|}$$

2 · CRITICAL MOMENTS: HOW FAR THE MOMENTS OF Y STAY FINITE

$$p^* = \sup\{p \geq 0 : \mathbb{E}[Y^{1+p}] < \infty\}, \quad q^* = \sup\{q \geq 0 : \mathbb{E}[Y^{-q}] < \infty\}$$

3 · LEE'S MOMENT FORMULA

$$\beta_R = \psi(p^*), \quad \beta_L = \psi(q^*), \quad \psi(p) = 2 - 4(\sqrt{p^2 + p} - p)$$

4 · LQD WITH EXPONENTIAL TAILS: READ OFF THE TAIL SCALES

$$p^* = \frac{1}{A_R} - 1, \quad q^* = \frac{1}{A_L}$$

- **Reading ψ :** it falls from $\psi(0) = 2$, only the mean finite, to 0 as $p \rightarrow \infty$: $\psi(1) = 0.34$, $\psi(16.9) = 0.029$. More finite moments, flatter wing. Inverse: $p = \frac{1}{2\beta} + \frac{\beta}{8} - \frac{1}{2}$.
- **In the app:** LQD slopes come from A_L , A_R , and $A_R < 1$ keeps $\beta_R < 2$. The default SVI chart keeps both slopes below 1.95; MCS reports its slopes and caps them at 1.95 only in its structural chart.

IN THE APP

Parametric ▶ Fit diagnostics ▶ wings · Lee · var-swap: A_L , A_R and both slopes. Compare shows Lee L/R per model; Quality flags a slope above 2.

SYMBOL	MEANING
$Y = S_T/F_T$	Terminal price over the forward; its mean is one
$k, w(k)$	Log-moneyness $\log(K/F)$; total implied variance $\sigma^2\tau$ at k
β_R, β_L	Right (call) and left (put) wing slopes of w ; always between 0 and 2
p^*	Right critical moment: $\mathbb{E}[Y^{1+p}]$ finite for $p < p^*$. The count starts after the mean, which is always finite
q^*	Left critical moment: $\mathbb{E}[Y^{-q}]$ finite for $q < q^*$; small when much probability sits near $Y = 0$
ψ	Lee's map from a critical moment to a wing slope
A_L, A_R	LQD tail scales (slide 14): $\mathbb{P}(X < x) \asymp e^{x/A_L}$, $\mathbb{P}(X > x) \asymp e^{-x/A_R}$, $X = \log Y$

RECORDED QQQ FIT · 18-DEC-26, 24 SEPTEMBER

Fit diagnostics: $A_L = 0.628$, $A_R = 0.056$. Left: $q^* = 1/0.628 = 1.59$, $\beta_L = \psi(1.59) = 0.243$. Right: $p^* = 1/0.056 - 1 = 16.9$, $\beta_R = \psi(16.9) = 0.029$. These are the card's Lee slopes: a steep put wing, a nearly flat call wing.

Variance swaps as an integrated constraint

A variance-swap quote is one number per node, but it prices the whole strike strip, wings included. The fit meets it through one extra least-squares row.

ONE NODE: $x = K/F$, LOG-RETURN $X = \log(S_T/F_T)$, VARIANCE CLOCK τ

1 · FAIR TOTAL VARIANCE: THE LOG CONTRACT, REPLICATED WITH OUT-OF-THE-MONEY OPTIONS

$$w_{VS} = 2 \left[\int_0^1 \frac{p(x)}{x^2} dx + \int_1^\infty \frac{c(x)}{x^2} dx \right] = -2 \mathbb{E}[X], \quad \sigma_{VS} = \sqrt{w_{VS}/\tau}$$

2 · THE OBJECTIVE AS THE SOLVER SEES IT: A PLAIN SUM OF SQUARED ROWS (PLUS THE MODEL'S PENALTY ROWS)

$$\min_{\theta} \sum_i (\sqrt{\lambda_i} r_i)^2 + r_{VS}^2, \quad r_{VS} = \sqrt{\lambda_{VS}} (\sigma_{VS}^{\text{model}} - \sigma_{VS}^{\text{quote}})$$

3 · THE WEIGHT OF THE VAR-SWAP ROW: A SHARE OF THE NODE'S OPTION WEIGHT

$$\lambda_{VS} = \frac{\text{pct}}{100} \sum_i \lambda_i, \quad \text{Hard pin: } \lambda_{VS} = 10^4 \sum_i \lambda_i$$

$c(x), p(x)$: undiscounted call and put in forward units. $\sigma_{VS}^{\text{quote}}$: the level on the card; $\sigma_{VS}^{\text{model}}$: the slice's own. r_i, λ_i : option residuals (vol units) and weights, slides 10 and 12. pct: the weight setting varSwapWeightPct, 10 by default.

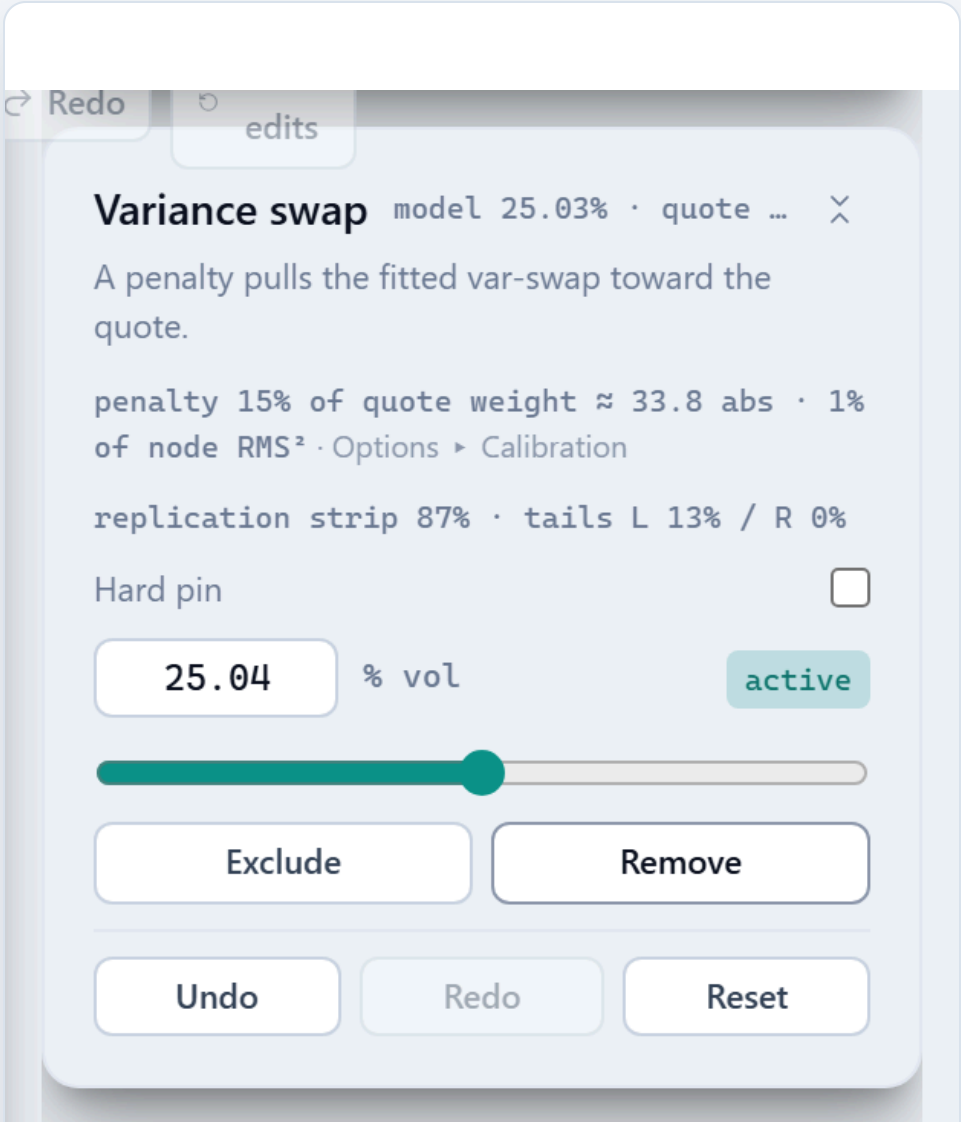
- **Why $\sqrt{\lambda}$:** the solver only squares its rows and adds them up. A row $\sqrt{\lambda} e$ squares to λe^2 , so λ is the weight of the miss e in the objective. Weights add up across rows, which is why the setting is a share of $\sum_i \lambda_i$.
- **Why a share:** 10% means the same on a 20-quote node and on a 200-quote node. At 100% the single quote weighs as much as all the options together.

RECORDED QQQ CARD · 24 SEPTEMBER

225 quotes at equal weight: $\sum_i \lambda_i = 225$. Saved setting 15%: $\lambda_{VS} = 0.15 \times 225 = 33.75$ (the card's 33.8), $\sqrt{\lambda_{VS}} = 5.81$. The model misses the quote by about 1 vol bp, so the row reads about 5.8 bp; squared, it weighs as much as 33.75 option quotes each 1 bp off.

IN THE APP

Parametric aside ▶ Variance swap card: add a quote at the model level, move it and watch the wings. The weight % sits in Options ▶ Calibration; Hard pin makes the row stiff.



QQQ 18-Dec-26, 24 September capture: quote 25.04%, model 25.03%. Weight 15% of the quote weight (default 10%), about 1% of the node's squared error. Replication: 87% from the quoted strikes, 13% from the put wing beyond them.

Calendar consistency across expiries

Two expiries can each be a valid law and still price a calendar spread below zero. The test compares calls at the same moneyness, or the LQD ledger at the same rank.

EXPIRIES $T_1 < T_2$, EACH NORMALISED BY ITS OWN FORWARD: $Y_T = S_T/F_T$, $\mathbb{E}[Y_T] = 1$

1 · SAME LOG-MONEYNESS k : CALLS IN FORWARD UNITS, EQUIVALENTLY TOTAL VARIANCE
$$c_{T_2}(k) \geq c_{T_1}(k) \quad \forall k \iff w_{T_2}(k) \geq w_{T_1}(k) \quad \forall k$$

2 · SAME RANK u : THE UPPER SHARE G OF SLIDE 15, CALLED THE LEDGER
$$G_T(u) = \int_u^1 e^{Q_T(v)} dv, \quad G_{T_2}(u) \geq G_{T_1}(u) \quad \forall u$$

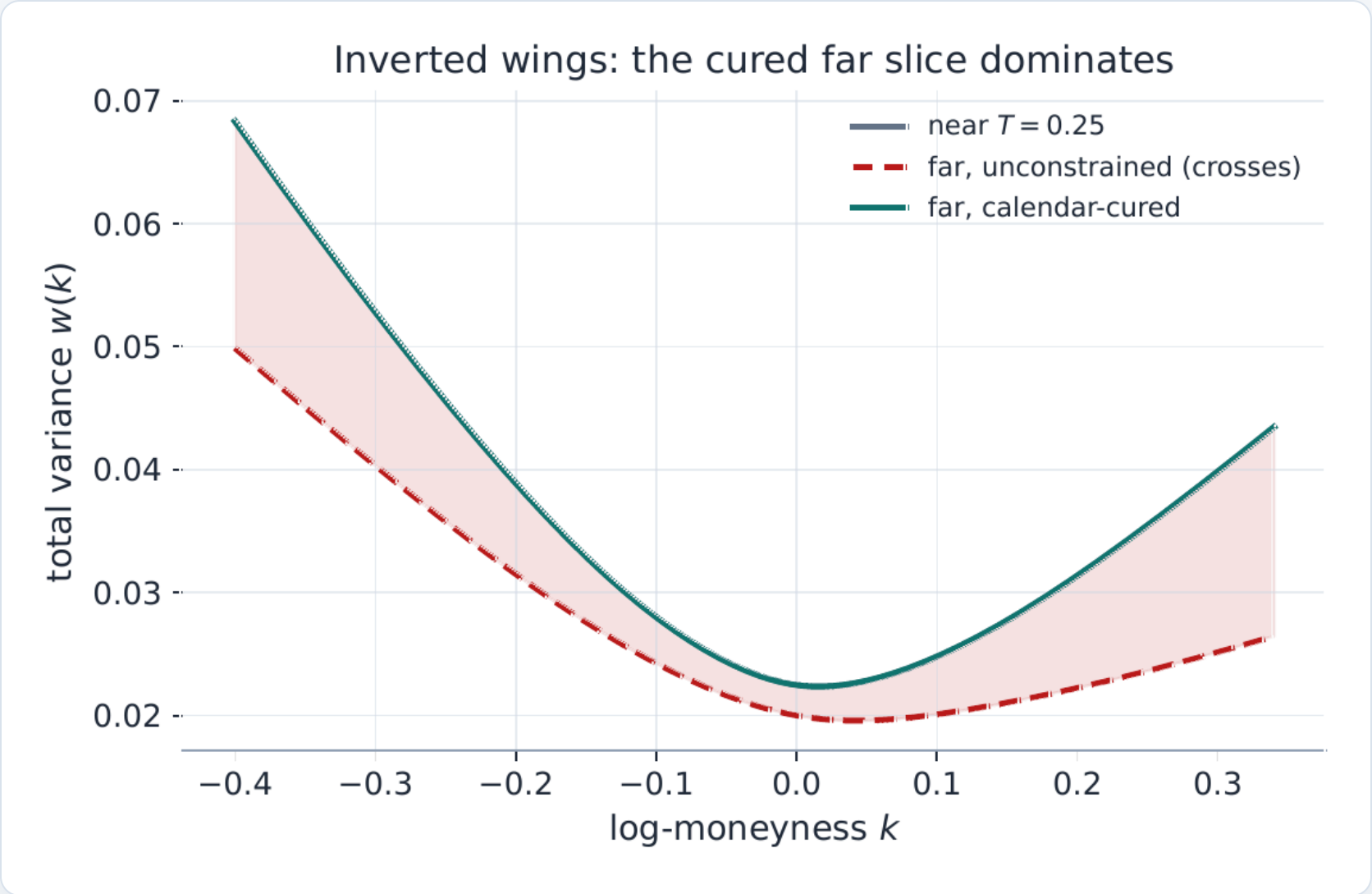
3 · WHY THE TWO TESTS AGREE: EACH CURVE IS AN EXTREMUM OVER THE OTHER
$$c_T(k) = \max_u [G_T(u) - e^k(1 - u)], \quad G_T(u) = \min_k [c_T(k) + e^k(1 - u)]$$

$c_T(k) = \mathbb{E}[(Y_T - e^k)^+]$: call in forward units. $w_T(k)$: total implied variance at $k = \log(K/F_T)$. Q_T : quantile of $\log Y_T$ at rank u . $G_T(u)$: the share of the forward paid by the outcomes above the u -quantile, from $G_T(0) = 1$ down to $G_T(1) = 0$.

- **Convex order:** equal means and ordered calls: the far law is a mean-preserving spread of the near one, and some martingale has both as marginals (Kellerer).
- **Variance, not vol:** the order is on w . At the money, $(w_{T_2} - w_{T_1})/(T_2 - T_1)$ is the forward variance; the vol itself may fall with maturity.

ILLUSTRATIVE CALCULATION

The figure's pair at the money: 3 months at 30%, $w = 0.0225$; 6 months at 20%, $w = 0.0200$. Forward variance $(0.0200 - 0.0225)/0.25 = -0.010$: an arbitrage. With 24% at 3 months ($w = 0.0144$), 20% at 6 months is fine: forward vol 15.0%.



Reference-note figure: total variance against k , near $T = 0.25$ (grey, under the teal), far $T = 0.5$. Fitted alone, the far slice (red, dashed) lies below the near one: negative forward variance across the shaded range. Refitted under a floor at the near slice (teal), it is lifted onto the near curve and no further.

Calendar repair and numerical certification

Fit each expiry alone, repair only the pairs that violate, then certify the stored surface on the whole line: at the nodes, between them and in the tails.

ADJACENT EXPIRIES NEAR AND FAR, ON THE LOG-ODDS GRID $z_j = -40 + j h$, $h = 0.01$

1 · LEDGER GAP AT THE SAME RANK, AND ITS EXACT SLOPE AT EVERY NODE

$$\Delta G(z) = G_{\text{far}}(z) - G_{\text{near}}(z), \quad \partial_z \Delta G = u(1 - u)(e^{Q_{\text{near}}(z)} - e^{Q_{\text{far}}(z)})$$

2 · HERMITE LEDGER GAP: ON EACH CELL, THE CUBIC THAT MATCHES BOTH END VALUES AND BOTH END SLOPES

$$\Delta \hat{G}(z_j + t h) = c_0 + c_1 t + c_2 t^2 + c_3 t^3, \quad 0 \leq t \leq 1$$

3 · INTERIOR CANDIDATES: WHERE THAT CUBIC'S SLOPE VANISHES

$$c_1 + 2c_2 t + 3c_3 t^2 = 0, \quad 0 < t < 1$$

4 · CERTIFICATE: THE MINIMUM OVER NODES, INTERIOR ROOTS AND ONE TAIL POINT PER SIDE

$$\min_{z \in \mathbb{R}} \Delta \hat{G}(z) \geq -10^{-6}$$

$G(z)$: the ledger of slide 23 at rank $u = 1/(1 + e^{-z})$; $Q_{\text{near}}, Q_{\text{far}}$: the two quantiles. With $d = \Delta G(z_{j+1}) - \Delta G(z_j)$, $m_0 = h \partial_z \Delta G(z_j)$, $m_1 = h \partial_z \Delta G(z_{j+1})$: $c_0 = \Delta G(z_j)$, $c_1 = m_0$, $c_2 = 3d - 2m_0 - m_1$, $c_3 = m_0 + m_1 - 2d$. Grid: 8,001 nodes, $|z| \leq 40$. Tolerance 10^{-6} of the forward, 0.01 bp.

- **Why exact:** the pricer reads the same Hermite cubics, so this minimum is the minimum of the stored surface, not a sample. A node-only check can miss a dip inside a cell (strip).
- **Tails:** beyond $|z| = 40$ both ledgers follow their tail continuations: one closed-form candidate per side. Tail order, far scales $\geq$ near ones, is a separate clause.

IN THE APP

Quality lens ▶ node card: Cal viol shows the screen and the certificate's ledger gap; Tail order reads the tail clause; a failed certificate blocks Publish.

STEP	CALCULATION	OUTCOME
1 · Independent fits	Each expiry on its own quotes, no calendar rows	A clean ladder keeps them, bit for bit
2 · Screen	Near minus far call on the common quote support, over vega	Violation above 0.5 vol bp
3 · Symmetric repair	Joint Gauss–Newton on each violating run; hinge weight $\times 10$, at most 3 times	Both slices concede, by their data weight
4 · Exchange	Where the certificate fails: stiff ledger row (10^6) at the worst rank; refit	Up to 8 rounds, else irreducible
5 · Certificate	Exact minimum of the Hermite ledger gap	Below -10^{-6} : not ready, Publish blocked
6 · Tail order	Far tail scales $\geq$ near ones, on each side	Advisory; blocks only with the Tail-order gate

ILLUSTRATIVE CALCULATION

One cell, $h = 0.01$: gap 2×10^{-6} at both nodes, slopes ∓ 0.002 . The cubic is $\Delta \hat{G} = 2 \cdot 10^{-6} (1 - 10t + 10t^2)$; its slope vanishes at $t = 1/2$, where $\Delta \hat{G} = -3 \times 10^{-6}$. Both nodes pass the -10^{-6} tolerance; the cell fails.

A piecewise-affine local-variance surface

The local-volatility fit has one unknown per vertex of a strike–time grid: the local variance there. Between vertices the surface is affine on triangles, so positive vertices give a positive surface.

LOCAL VARIANCE $\nu(\tau, x)$ AT NORMALISED STRIKE $x = K/F$ AND VARIANCE TIME τ

1 · ONE UNKNOWN PER VERTEX, ONE HAT FUNCTION PER UNKNOWN

$$\nu_{\theta}(\tau, x) = \sum_{\ell=1}^m \theta_{\ell} \phi_{\ell}(\tau, x), \qquad \theta_{\ell} = \nu_{\theta}(\tau_{\ell}, x_{\ell})$$

2 · INSIDE A TRIANGLE: AN AVERAGE OF ITS THREE CORNERS

$$\nu_{\theta}(\tau, x) = b_1 \theta_{\ell_1} + b_2 \theta_{\ell_2} + b_3 \theta_{\ell_3}, \qquad b_a \geq 0, \quad b_1 + b_2 + b_3 = 1$$

3 · SO A BOX ON THE VERTICES IS A BOX ON THE WHOLE SURFACE

$$\nu_{\text{lo}} \leq \theta_{\ell} \leq \nu_{\text{hi}} \quad \text{for every } \ell \quad \implies \quad \nu_{\text{lo}} \leq \nu_{\theta}(\tau, x) \leq \nu_{\text{hi}}$$

ϕ_{ℓ} : hat function of vertex (τ_{ℓ}, x_{ℓ}) , 1 there, 0 at every other vertex, affine on each triangle. $m = n_{\tau} n_x$ vertices; each grid rectangle is cut into two triangles. b_a : barycentric weights of (τ, x) in its triangle, with corners ℓ_1, ℓ_2, ℓ_3 .

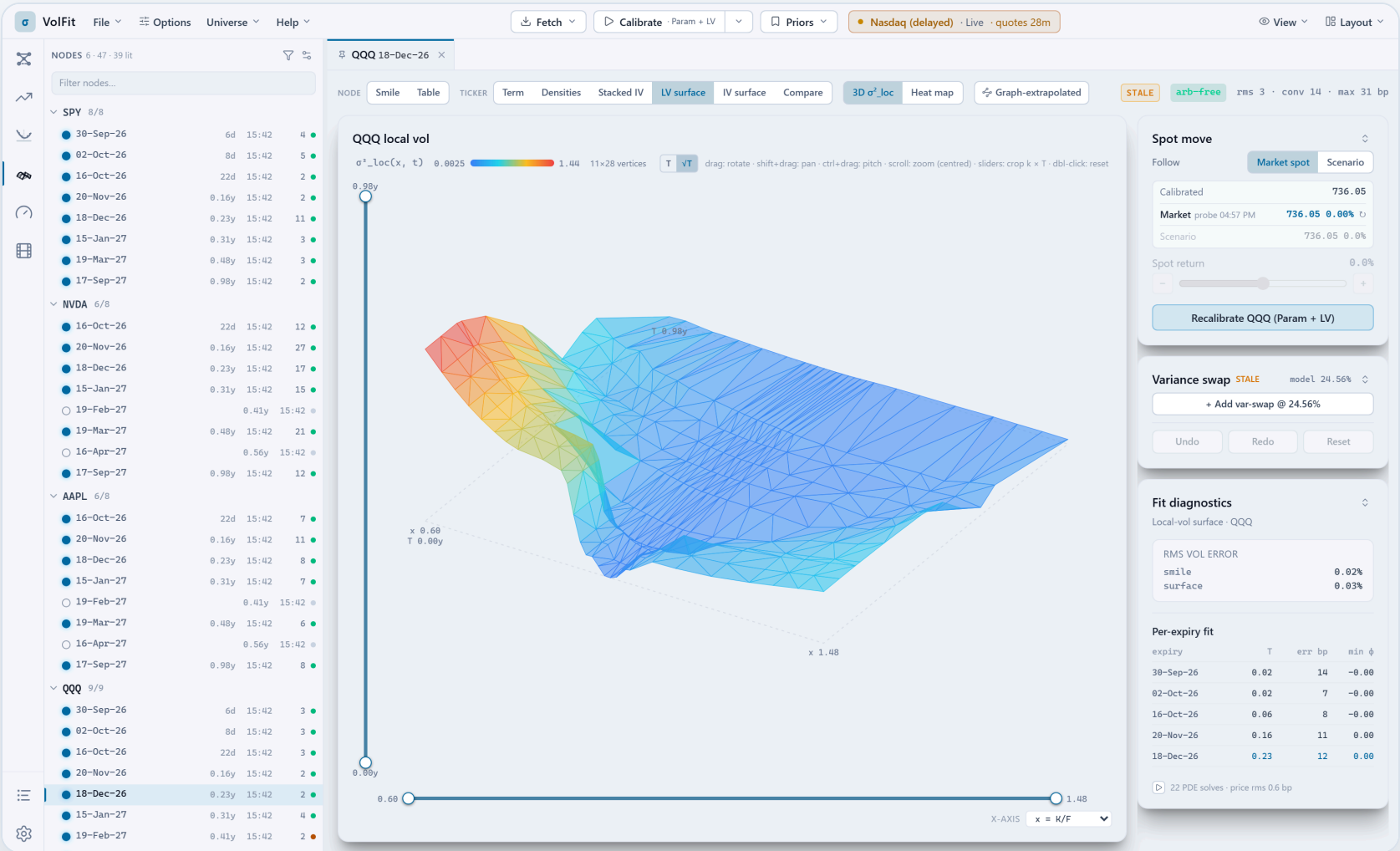
- **Not an implied vol:** a price averages local variance along the paths to expiry; at the money, $\sigma^2_{\text{ATM}}(\tau) \approx \frac{1}{\tau} \int_0^{\tau} \nu(s, 1) ds$. A vertex is read through the pricing equation.
- **Vertices:** time rows at 0, $\tau_1/4$ and each expiry, at least 10; strike columns on a delta axis in the quoted range, at least 12, and 8 inside each expiry's range.
- **Outside the grid:** flat on the right. On the left flat by default, or linear at a times the first cell's slope with Convex wing ($a = 1.5$) or a variance-swap quote (a fitted).

WORKED EXAMPLE

Weights 0.5, 0.3, 0.2 on corners at 20%, 30%, 25% local vol:
 $\nu = 0.5(0.04) + 0.3(0.09) + 0.2(0.0625) = 0.0595$, a local vol of 24.4%. Variances are averaged, not vols, and the result stays between 20% and 30%.

IN THE APP

Local Vol lens (Alt+4) ▶ LV surface: rotate the mesh. Options ▶ Local-Vol surface: the resolved grid $n_{\tau} \times n_x = m$ and the box.



24 September capture of the fitted QQQ sheet: 11×28 vertices, time rows by strike columns. Each facet is a triangle on which ν is affine; height and colour give the local variance, 0.0025 to 1.44. STALE refers to the saved snapshot.

The forward Dupire pricing calculation

Given a local-variance sheet, one forward march prices every strike and every expiry: start from the payoff and step forward in time, one tridiagonal solve per step.

CALL OVER DF , $c(\tau, x)$, AT NORMALISED STRIKE $x = K/F$; ON THE LATTICE, $U_i^n = c(\tau_n, x_i)$

1 · FORWARD DUPIRE EQUATION, STARTED FROM THE PAYOFF

$$\partial_\tau c = \frac{1}{2} \nu(\tau, x) x^2 \partial_{xx} c, \quad c(0, x) = (1 - x)^+$$

2 · THREE-POINT STENCIL: ROW i OF THE MATRIX A

$$(AU)_i = \nu_i \left[a_i^- U_{i-1} - (a_i^- + a_i^+) U_i + a_i^+ U_{i+1} \right]$$

3 · ONE BDF2 STEP: A SINGLE TRIDIAGONAL SOLVE

$$(I - \gamma_n \Delta \tau_n A^{n+1}) U^{n+1} = \alpha_n U^n - \beta_n U^{n-1}$$

4 · STEP WEIGHTS FROM THE STEP RATIO $\omega_n = \Delta \tau_n / \Delta \tau_{n-1}$

$$\gamma_n = \frac{1 + \omega_n}{1 + 2\omega_n}, \quad \alpha_n = \frac{(1 + \omega_n)^2}{1 + 2\omega_n}, \quad \beta_n = \frac{\omega_n^2}{1 + 2\omega_n}$$

Boundaries: $c(\tau, 0) = 1$ and $c(\tau, x_{\max}) = 0$, with $x_{\max} = \max(1.4 x_{\text{top}}, 2.5)$ and x_{top} the highest quoted strike. Step one, and any step more than twice the previous one, is implicit Euler: $(\gamma, \alpha, \beta) = (1, 1, 0)$.

- What BDF2 is: backward differentiation, order 2: the time derivative at τ_{n+1} is the slope of the parabola through U^{n-1} , U^n and U^{n+1} , with A taken at τ_{n+1} . Error $O(\Delta \tau^2)$; the stiffest modes are damped to zero (L-stable), so the payoff kink does not ring.
- Grids: strike step $0.15 \sigma \sqrt{\tau}$ over each expiry's range $\pm 6 \sigma \sqrt{\tau}$, 0.02 in the wings, $x = 1$ a node. Time steps start at 1% of the first grid mark and grow by at most 25%, capped at 0.05; every expiry and vertex row is on the grid.

SYMBOL	MEANING
c, ν	Call over DF , carry inside F and D ; local variance, the sheet of the previous slide
τ	Variance time (slide 29): calendar time plus event days; t itself with no events
$x_i, h_i^\pm$	Lattice node i ; widths of the cells below (h_i^-) and above (h_i^+) it
$\nu_i, a_i^\pm$	$\nu(\tau_{n+1}, x_i)$; stencil weights $x_i^2 / ((h_i^- + h_i^+) h_i^\pm)$, the $\frac{1}{2}$ folded in
U^n	Call values on the lattice at time level τ_n
$\Delta \tau_n, \omega_n$	Step $\tau_{n+1} - \tau_n$; its ratio to the previous step
$\gamma_n, \alpha_n, \beta_n$	Step weights: implicit Euler $(1, 1, 0)$; constant-step BDF2 $(\frac{2}{3}, \frac{4}{3}, \frac{1}{3})$

WORKED NUMBERS

A flat 15% sheet, whose exact price is known, marched over the SPY ladder of 23 September (2 to 359 days): 115 graded steps, the first 7 minutes long. ATM vol error: implicit Euler 16 bp at 2 days, 5 to 12 bp further out; BDF2 3.2 bp at 2 days, at most 0.3 bp further out.

Calibrating through the PDE

The fit chooses the vertex values θ by damped Gauss–Newton. The derivatives it needs come out of the same march as the prices, through the same tridiagonal matrix.

UNKNOWN θ : THE m VERTEX VARIANCES; QUOTES $i = 1, \dots, n$

1 · QUOTE RESIDUAL: THE PRICE MISS OVER VEGA, IN VOL POINTS
$$r_i(\theta) = \frac{c_i(\theta) - c_i}{0.01 v_i}$$

2 · OBJECTIVE: WEIGHTED MISSES PLUS THREE PENALTIES, MINIMISED OVER THE BOX $\nu_{lo} \leq \theta \leq \nu_{hi}$
$$\Phi(\theta) = \sum_i \lambda_i r_i^2 + \kappa_R \|\delta^2 \theta\|^2 + \kappa_D \|\delta_x^3 c\|^2 + \kappa_F \|\theta_{0,\cdot} - \theta_{1,\cdot}\|^2$$

3 · SENSITIVITIES $S_\ell = \partial U / \partial \theta_\ell$: THE SAME MATRIX, ONE MORE RIGHT-HAND SIDE EACH
$$(I - \gamma \Delta \tau A^{n+1}) S_\ell^{n+1} = \alpha S_\ell^n - \beta S_\ell^{n-1} + \gamma \Delta \tau A_\ell^{n+1} U^{n+1}$$

4 · LEVENBERG–MARQUARDT STEP ON THE STACKED RESIDUALS r , WITH $\|r\|^2 = \Phi$, CLIPPED TO THE BOX
$$(J^\top J + \mu \operatorname{diag}(J^\top J)) \Delta \theta = -J^\top r, \quad \theta \leftarrow \operatorname{clip}(\theta + \Delta \theta)$$

- **Why tridiagonal:** the second difference at node i uses nodes $i - 1, i, i + 1$ only: three entries per row of A . One pivot-free factorisation per step, $O(N_x)$, serves the price and all m columns S_ℓ .
- **Gauss–Newton, Levenberg–Marquardt:** linearise, $r(\theta + \Delta \theta) \approx r + J \Delta \theta$, and solve that least squares. The damping μ shortens the step toward the gradient: it falls after good steps, rises after rejected ones. LSMR needs only Jv and $J^\top w$.
- **Regularisation:** prices average local variance over paths, so many sheets fit the same quotes. The penalties pick one: least curvature, smoothest density, and early rows tied to the first expiry, which pins only their average.

IN THE APP Local Vol aside ▶ Fit diagnostics: per-expiry errors, N PDE solves · price rms, and the ▶ calibration-trace player.

SYMBOL	MEANING
$c_i(\theta), c_i$	Model call at quote i , read off the march; the market mid. Both over DF
v_i, λ_i	Black vega of the quote, floored at 10^{-3} ; quote weight. A 1-point vol miss gives $r_i \approx 1$
$\delta^2 \theta$	Second differences of neighbouring vertices along strike and along time, spacing-aware: $\theta_{j-1} - 2\theta_j + \theta_{j+1}$ on an even grid
$\delta_x^3 c$	Third differences of each expiry's marched call, the slope of the density $\partial_{xx} c$, over the quoted range ± 2 ATM std
$\theta_{0,\cdot}, \theta_{1,\cdot}$	Vertex rows at $\tau = 0$ and the next one; before a front under 0.08 y, every row up to it, at weight at least 1
$\kappa_R, \kappa_D, \kappa_F$	Penalty weights, by default $10^{-2}, 1, 10^{-2}$ (Options: Roughness, Density smoothness, Front-tie weight)
A_ℓ	$\partial A / \partial \theta_\ell$: A with ν_i replaced by $\phi_\ell(\tau_{n+1}, x_i)$; γ, α, β as on slide 26
J, μ	Jacobian of all rows of r , built from the S_ℓ ; damping, 10^{-3} at the start

RECORDED SPY · 23 SEPTEMBER

875 quotes, $m = 253$ vertices, lattice 368 strikes by 115 steps. One evaluation marches the price and up to 253 columns S_ℓ : $115 \times 366 \times 253 \approx 10.6$ million node updates behind one factorisation per step. Cold fit: 66 evaluations, 2.8 s. Warm-started from the last fit: 20 evaluations, 0.9 s.

Fitted local variance and the Dupire-derived surface

The Compare tab reads a local variance off the parametric smiles the classical way, differentiate then divide, and scores it beside the fitted sheet on the same quotes.

ONE EQUATION, READ IN BOTH DIRECTIONS

1 · DUPIRE TWIN: DIFFERENTIATE THE PARAMETRIC SURFACE, THEN DIVIDE

$$\nu_{\text{twin}}(\tau, x) = \frac{\partial_{\tau} c}{\frac{1}{2} x^2 \partial_{xx} c} = \frac{\partial_{\tau} w}{g(k, w)}, \quad k = \log x$$

2 · ROUND TRIP: REPRICE THE TWIN, COMPARE IT WITH ITS OWN SOURCE AT THE QUOTED k_i

$$\text{RT} = \text{rms}_i \left| \sigma_{\text{twin}}(k_i) - \sigma_{\text{param}}(k_i) \right|$$

$w(k, \tau)$: the parametric total variance, interpolated in τ by a monotone cubic (Smooth) or linearly (Buckets).
 $g = \left(1 - \frac{k \partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{1}{2} \partial_{kk} w$, the butterfly function; cells with $g \leq 0$ or $\partial_{\tau} w \leq 0$ are counted as repairs.
 $\sigma_{\text{twin}}, \sigma_{\text{param}}$: implied vols of the repriced twin and of its source.

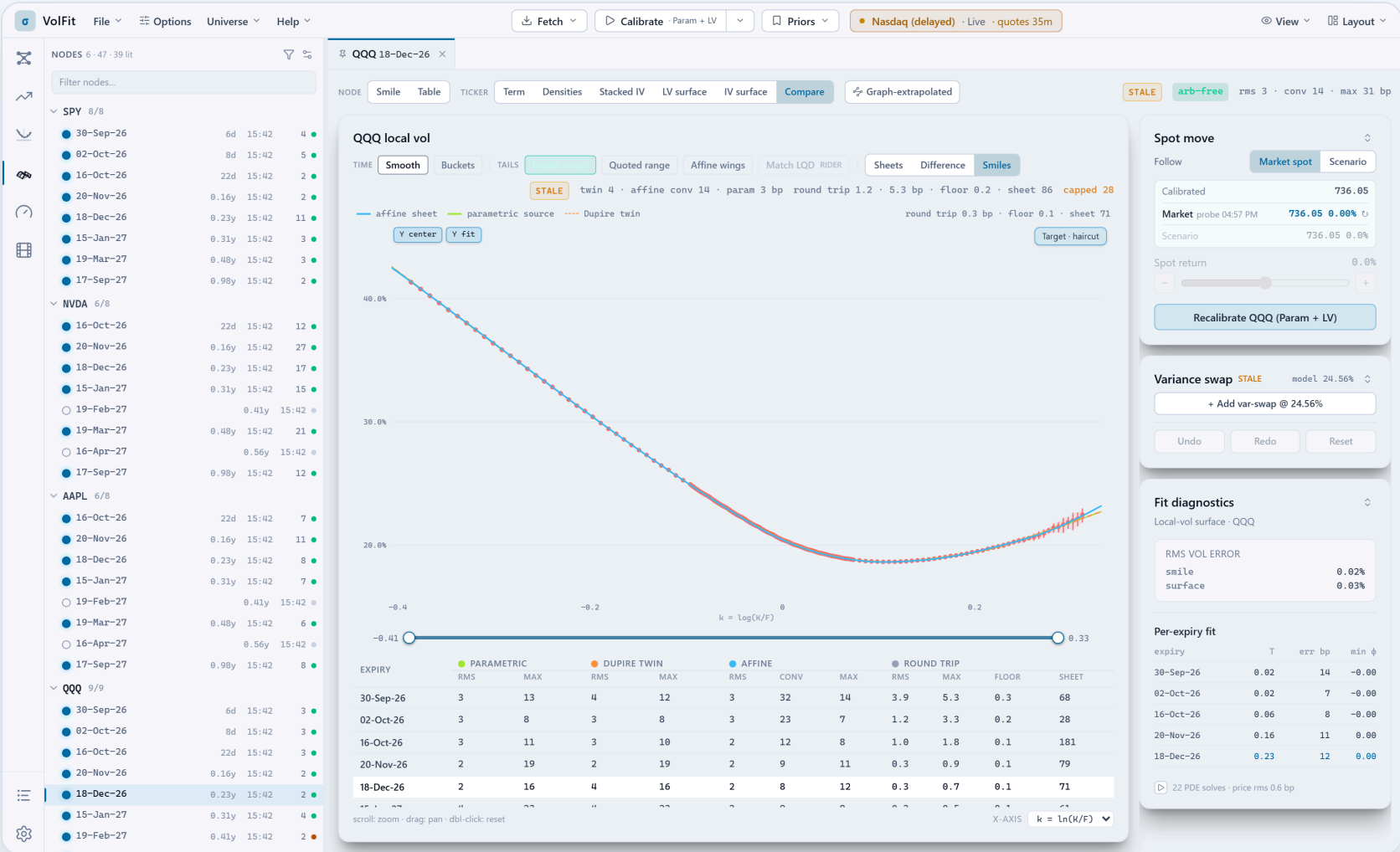
- **Twin**: repriced on the fit's lattice made four times finer in strike and eight times finer in time, second-order (Rannacher). Read-only: the tab never writes to the fit.
- **Affine**: the direct fit of slides 25–27. rms: on its own lattice; conv: the same sheet repriced on a lattice twice as fine in strike and four times in time.
- **Controls**: floor: a flat sheet through the twin's operator, whose error is the operator's own. sheet: the twin sampled at the fit's vertices and repriced, which is what the vertex grid loses.

RECORDED SPY · 23 SEPTEMBER

Nine expiries, 875 quotes, 253 vertices: 3.61 vol bp RMS on the fit's own lattice, 5.12 bp repriced on the finer one, 24.8 bp worst quote. Quote both: a fit can absorb its own lattice's error into θ , and only another lattice shows it.

IN THE APP

Local Vol ▶ Compare ▶ Smiles: the three curves on the quotes and the per-expiry score table. The Time and Tails chips rebuild the twin; Sheets and Difference set the two surfaces on one vertex grid.



24 September capture, QQQ 18-Dec-26, haircut target: affine sheet (blue), parametric source (green), Dupire twin (orange, dashed). Its row reads parametric 2, twin 4, affine 2 (conv 8) bp; round trip 0.3 bp over a floor of 0.1, while the twin sampled at the vertices reprices 71 bp away.

Event-weighted variance time

A price fixes total variance w ; a volatility is w divided by a clock. The event clock counts a scheduled event as extra days, so the same price reads a lower volatility.

EXPIRY AT CALENDAR TIME t IN YEARS; SCHEDULED EVENTS e ON DATES t_e

1 · VARIANCE CLOCK: EACH EVENT BEFORE EXPIRY ADDS N_e DAYS

$$\tau(t) = t + \frac{1}{365} \sum_{t_e \leq t} N_e$$

2 · TWO READINGS OF ONE PRICE

$$w = \sigma_{\text{cal}}^2 t = \sigma_{\text{evt}}^2 \tau \implies \sigma_{\text{cal}} = \sigma_{\text{evt}} \sqrt{\tau/t}$$

3 · OPTIONAL NORMALISATION: A ONE-YEAR EXPIRY KEEPS A ONE-YEAR CLOCK

$$\tau_{\text{norm}}(t) = \tau(t) \frac{365}{365 + \sum_{t_e \leq 1} N_e}$$

N_e : extra equivalent days of event e ($N_e = 4$: that day counts as five). σ_{evt} : the working volatility read by every fit and view. Carry, discounting and early exercise stay on calendar time t .

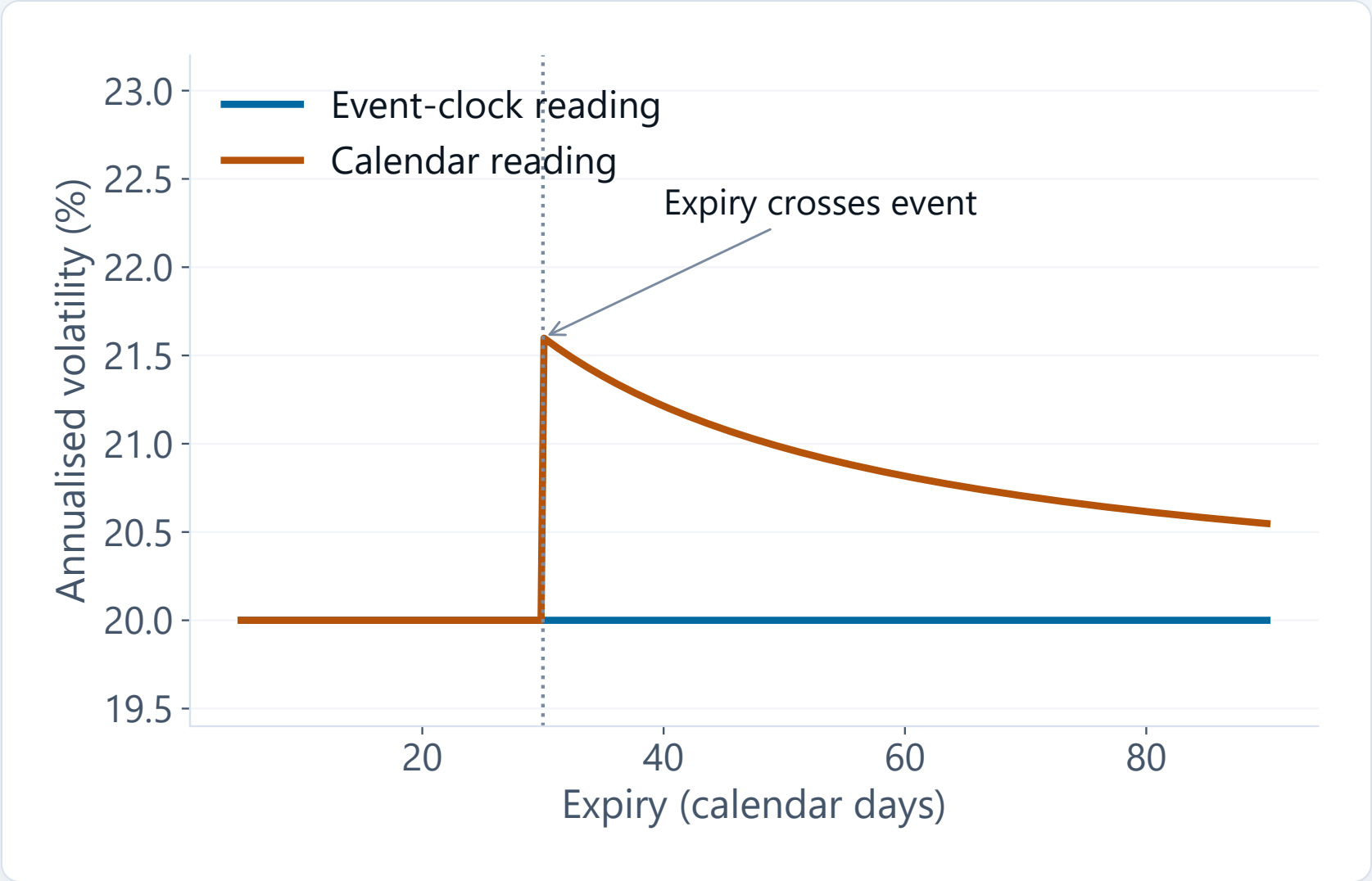
- Only the reading changes: prices, total variance and the butterfly and calendar checks are the same on both clocks. With no events, $\tau = t$ exactly.
- After the event: its N_e leaves the remaining horizon. The calendar reading falls back to the event-clock level, the vol crush; the event-clock reading does not move.

WORKED EXAMPLE

Event on day 30 worth $N_e = 5$, 20% on the event clock. 30-day expiry: $\tau = 35/365$, $w = 0.04 \times 35/365 = 0.00384$, calendar reading $20\% \times \sqrt{35/30} = 21.60\%$. At 60 and 90 days: 20.82% and 20.55%. Normalised: every τ is multiplied by $365/370$, and the same prices read 20.14% at every expiry.

IN THE APP

Parametric ▶ Term: the Events card (date in years, extra days). The chart draws the event-clock reading and, dashed amber, the calendar reading of the same prices. Options ▶ Events and Normalize events.



Same prices, two clocks: the event-clock reading is flat at 20%; the calendar reading jumps to 21.60% when expiry crosses the day-30 event, then decays as the event becomes a smaller share of the horizon.

Auto-calibrating isolated variance peaks

Prices fix the forward variance between consecutive expiries. An interval that runs hotter than both neighbours is read as one event, sized exactly to remove the excess; every other shape installs nothing.

INTERVAL i RUNS FROM EXPIRY $i - 1$ TO EXPIRY i AND SPANS d_i CALENDAR DAYS

1 · FORWARD VARIANCE PER CALENDAR YEAR: FIXED BY PRICES

$$f_i = \frac{w_i - w_{i-1}}{t_i - t_{i-1}}$$

2 · REFERENCE: THE HOTTER NEIGHBOUR, READ ON THE EVENT CLOCK

$$r_i = \max(f_{i-1}^\tau, f_{i+1}^\tau), \quad f_j^\tau = \frac{w_j - w_{j-1}}{\tau_j - \tau_{j-1}}$$

3 · EVENT SIZE: THE EXTRA DAYS THAT BRING f_i DOWN TO r_i

$$N_i = d_i \left(\frac{f_i}{r_i} - 1 \right)$$

4 · MATERIALITY: BOTH FLOORS, OR NO EVENT

$$N_i \geq \max(0.5, 0.03 d_i) \text{ days}$$

w_i : ATM total variance of expiry i from its fitted slice, the same on both clocks. t_i, τ_i : calendar and event-clock times; f_j^τ includes the events already found, so the rule repeats until no peak is left.

- **Peaks only:** an event can only lower its own interval's event-clock rate. A ramp, a smooth backwardation or a dip cannot come from an event: nothing is installed.
- **Ends and placement:** the last interval is never a candidate; the first compares with its right neighbour, extended by the back ladder's slope when that ladder falls. An event goes at its interval's midpoint; edit the date if you know it.

INTERVAL (DAYS)	ATM VOL AT ITS END, CALENDAR	FORWARD VOL $\sqrt{f_i}$	FOUND N_i
0–30	20.00%	20.00%	0
30–60	20.66%	21.29%	4.0 days, at day 45
60–90	20.44%	20.00%	0
90–120	20.33%	20.00%	reference only

WORKED EXAMPLE

Monthly ladder, 20% on the event clock, earnings worth 4 extra days between days 30 and 60. That interval is hotter than both neighbours: $N = 30 (0.04533/0.04 - 1) = 4.0$ days. The Ladder spread readout goes from 53 variance bp on the calendar clock to 0 on the event clock.

Spot transport between calibrations

Between calibrations the stored fit is read through a transport, not refitted; the spot–vol rule R of slide 5 sets how far the smile slides.

FORWARD MOVE $h = \log(F_1/F_0)$; R IS THE SPOT–VOL RULE

1 · THE NEW CURVE READS THE STORED ONE Rh FURTHER ALONG; A FIXED STRIKE'S LABEL MOVES BY $-h$

$w_1(k) = w_0(k + Rh), \quad k_1 = k_0 - h \implies w_1(k_1) = w_0(k_0 + (R - 1)h)$

2 · ATM RESPONSE, TO FIRST ORDER

$\Delta\sigma_{\text{ATM}} = \sigma_0(Rh) - \sigma_0(0) \simeq R s_0 h$

3 · STICKY LOCAL VOL: HAGAN'S MAP, $w_1(k) = w_0(\ell(k, h))$

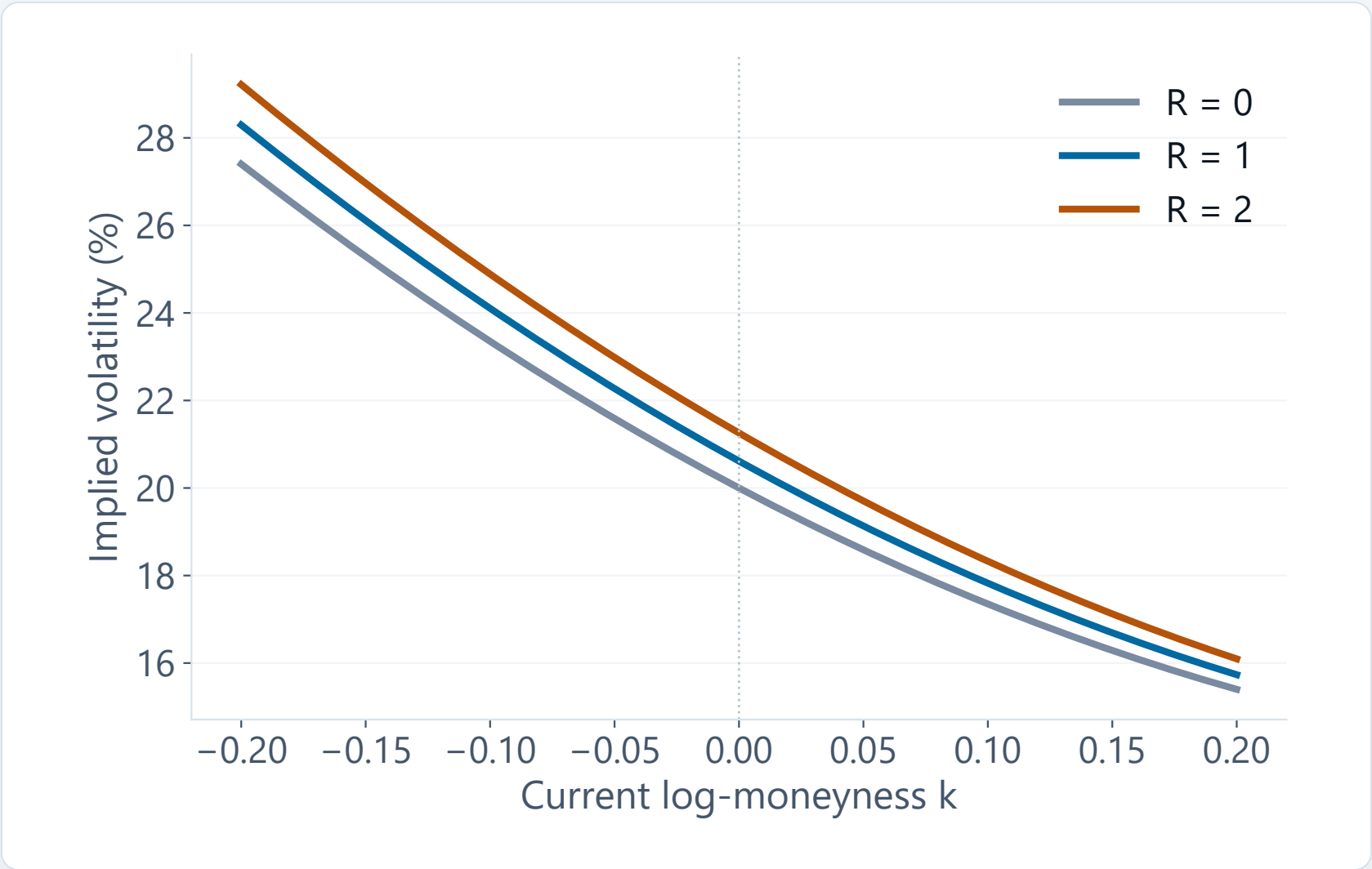
$\ell(k, h) = \log(e^h(e^k + 1) - 1) \simeq k + (1 + e^{-k})h$

F_0, F_1 : stored and current forward; w_0, σ_0 : stored variance and vol; $s_0 = \partial_k \sigma_0(0)$: ATM skew; τ is unchanged.

- **The dial R :** 0 sticky moneyness, the smile rides the forward; 1 sticky strike, the default: every strike keeps its vol; 2 sticky local vol: at the money the curve slides twice as far, $\ell(0, h) \simeq 2h$.
- **Hagan's map:** in a local-vol model the implied vol at K is, to leading order, the local vol at the midpoint $(F + K)/2$. Freeze the local-vol surface in strike: the new smile at K reads the stored one at the strike with the same midpoint against F_0 , which is ℓ .
- **What is recomputed:** nothing: every view reads the stored fit through the map, no PDE; only the Scenario overlay's LV-grid regime reprices through Dupire, on a grid taken from the parametric smiles.

WORKED EXAMPLE

The new ATM reads the stored curve at Rh . $R = 1$: $\sigma_0(-0.02) = 20.61\%$, +61 bp (first order +60). $R = 2$: $\sigma_0(-0.04) = 21.26\%$, +126 bp; Hagan's map, $\ell(0, h) = -0.0404$, gives +127 bp.



Three readings of one stored smile: $\sigma_0(k) = 0.20 - 0.30k + 0.35k^2$ after a 2% forward fall, $h = -0.02$. Each curve is the stored one shifted right by $0.02R$; with a negative skew the ATM vol rises with R .

Activating a prior where quote support is weak

Yesterday's smile may pull on today's fit only where today's quotes leave a feature undetermined. Each persisted feature has a gate: fully open with no nearby quote, exactly closed from about one quote on.

PER EXPIRY: OPERATORS ATM, RR25, BF25 WITH LEGS k_a ON THE TRANSPORTED PRIOR

1 · SUPPORT OF A LEG: EFFECTIVE NUMBER OF QUOTES NEAR IT

$$n_a = \sum_i \tilde{\lambda}_i \exp\left(-\frac{(k_i - k_a)^2}{2b^2}\right)$$

2 · INFORMATION OF AN OPERATOR, $I_{\text{ATM}} = n_0$: EVERY LEG MUST BE SUPPORTED

$$I_{\text{RR}} = \left(\frac{1}{n_c} + \frac{1}{n_p}\right)^{-1}, \quad I_{\text{BF}} = \left(\frac{1}{4n_c} + \frac{1}{4n_p} + \frac{1}{n_0}\right)^{-1}$$

3 · GATE: 1 WITH NO SUPPORT, EXACTLY 0 FROM THE REQUIRED LEVEL

$$g_j = \left[\text{clip}\left(1 - \frac{I_j}{I_{\text{req}}}, 0, 1\right)\right]^\gamma$$

4 · ONE ROW PER OPEN GATE; ROWS SHARE A BUDGET

$$r_j = \sqrt{\Lambda_j} \left(O_j(\theta) - O_j(\text{prior})\right), \quad \Lambda_j = B \frac{g_j}{\sum_m g_m}$$

- **Order:** transport the saved prior to today's forward (previous slide), place the legs on it once, measure support from today's quotes, then gate.
- **A closed gate is exact:** $g_j = 0$ gives $\Lambda_j = 0$: the row adds no cost and no gradient, so the prior cannot pull on a feature the quotes pin down. Support counts quotes; it does not check that they agree.

IN THE APP

Options ▶ Prior ▶ Config: the Prior diagnostics table lists, per expiry, the open operators with gap · λ · age · src · h . An operator the quotes pin down drops out of the table.

SYMBOL	MEANING
$k_i, \tilde{\lambda}_i$	Log-moneyness of quote i ; its fit weight rescaled to mean one
n_0, n_c, n_p	Support at the ATM leg and at the 25Δ call and put legs
b	Kernel bandwidth: 0.06 in log-moneyness
I_j, I_{req}	Information of operator j ; required level 1, one effective quote
g_j, γ	Gate, the gap column in the app; sharpness, 1
$O_j(\theta), O_j(\text{prior})$	The operator, in vol, on the model's smile and on the transported prior
Λ_j, B	Weight of prior row j (column λ in the app); budget $B = 50\% \times \sum_i \lambda_i$
g_{VS}	Gate of the var-swap level: its own row, weight $B g_{\text{VS}}$, support probed at 0 and $\pm 1.4 \sigma_{\text{ATM}} \sqrt{\tau}$

WORKED EXAMPLE

One-year node, flat 20% prior, five equal-weight quotes at $k = 0, \pm 0.02, \pm 0.04$:
 $B = 2.5$. Supports $n_0 = 4.49, n_c = 0.30$ ($k = 0.155$), $n_p = 1.02$ ($k = -0.115$).
ATM closed. RR: $I = 0.23, g = 0.77, \Lambda = 1.91$. BF: $I = 0.76, g = 0.24, \Lambda = 0.59$.
Var swap: $g = 1$, weight 2.5.

Persisting shape versus absolute strike values

What the prior persists decides which moves it resists: risk reversals and butterflies ignore a parallel shift of the smile; a fixed-strike anchor does not.

LEGS AT THE 25Δ CALL AND PUT STRIKES k_c, k_p , FROZEN ON THE TRANSPORTED PRIOR

1 · SHAPE OPERATORS: COEFFICIENTS THAT SUM TO ZERO

$$RR = \sigma(k_c) - \sigma(k_p), \quad BF = \frac{1}{2}(\sigma(k_c) + \sigma(k_p)) - \sigma(0)$$

2 · SO A PARALLEL MOVE BY v VOL POINTS LEAVES BOTH UNCHANGED

$$RR(\sigma + v) = RR(\sigma), \quad BF(\sigma + v) = BF(\sigma)$$

3 · STRIKE ANCHOR: A PRICE RESIDUAL OVER VEGA, READ IN VOL

$$r_j = \sqrt{\Lambda_j} \frac{c_\theta(k_j) - c_{\text{prior}}(k_j)}{\partial_\sigma B(k_j) + \eta} \simeq \sqrt{\Lambda_j} (\sigma_\theta(k_j) - \sigma_{\text{prior}}(k_j))$$

$k_\Delta = \frac{1}{2}\sigma^2\tau - \sigma\sqrt{\tau}N^{-1}(\Delta)$: strike of forward call delta Δ (0.25 call, 0.75 put). c : call price in forward units. Λ_j : row weight (previous slide).

- **Hybrid, the default:** ATM, RR25, BF25 and the var-swap level carry shape where their gates open; anchors at the 2Δ, 5Δ and 10Δ strikes hold the deep tail no operator reaches. The anchor rows share a budget of 20% of the node's summed quote weights $\sum_i \lambda_{i\cdot}$ split by each anchor's missing quote mass: zero where quotes already cover it.
- **Across a jump:** a shape row follows today's level; a strike anchor pulls the unquoted wing back toward yesterday's absolute vol.

WORKED EXAMPLE

Yesterday ATM 20%, 25Δ call 19%, put 23%: $RR = -4$, $BF = +1$ vol point. Overnight every vol rises 4 points: 24%, 23%, 27%. RR and BF are unchanged, so their rows cost nothing. A strike anchor there, as in strike-gap mode, still targets 23%: a 4-point residual pulling the wing down.

IN THE APP

On a thin node with a saved prior: the chart header's Fit switch, Production (the + Prior fit) against Free; Parametric ▶ Compare shows the Pull column in ATM vol bp.



Illustrative: the whole smile lifted by 4 points keeps its RR and BF at fixed legs. A fit anchored to the earlier absolute wings keeps the old level outside the quoted band (shaded).

Temporal filtering of ATM handles

Each calibration is a noisy reading of three numbers: ATM level, skew and curvature. The filter carries yesterday's estimate to today and averages it with today's reading, each weighted by the inverse of its variance.

PER NODE, ONE SCALAR FILTER FOR EACH HANDLE $x \in \{\sigma_0, s_0, \kappa_0\}$

1 · PREDICT: MOVE THE LAST ESTIMATE TO TODAY'S FORWARD; THE WAIT ADDS VARIANCE

$$m^- = \mathcal{T}_h(m_{\text{prev}}^+), \quad P^- = P_{\text{prev}}^+ + Q$$

2 · MEASURE: TODAY'S FIT READS THE HANDLE AS z , WITH A VARIANCE R BUILT ON THE NEXT SLIDE

$$z = x(\hat{\theta}), \quad R = \text{Var}(z)$$

3 · UPDATE: THE INVERSE-VARIANCE AVERAGE OF PREDICTION AND READING

$$m^+ = \frac{Rm^- + P^-z}{P^- + R} = m^- + K(z - m^-), \quad K = \frac{P^-}{P^- + R}, \quad \frac{1}{P^+} = \frac{1}{P^-} + \frac{1}{R}$$

4 · THE TRANSPORT OF THE THREE HANDLES, FIRST ORDER IN THE FORWARD MOVE h

$$\sigma'_0 = \sigma_0 + \text{SSR } s_0 h, \quad s'_0 = s_0 + \kappa_0 h, \quad \kappa'_0 = \kappa_0$$

- **Never a setting:** the gain follows from the two variances. A precise reading, small R , moves the estimate almost fully; a wide one barely.
- **Per handle:** the three updates are separate: on coarse chains, correlations between level and curvature let a junk curvature reading drag the level by 3 to 28 vol points.

IN THE APP

Options ▶ Kalman filter: Observation filter Overlay; the table K(ATM) · K(skew) · K(curv) · innov bp, then Timeline. On the smile, the FILTER badge.

SYMBOL	MEANING
σ_0, s_0, κ_0	ATM vol, skew $\partial_k \sigma(0)$ and curvature $\partial_k^2 \sigma(0)$ of the LQD fit (slide 17)
m^-, P^-	Predicted value and variance of a handle, before today's reading
m^+, P^+	Its filtered value and variance after the reading
z, R	Today's fitted handle and its variance, from the fit itself (next slide)
Q	Variance added by the wait: $(30 \text{ bp})^2$ per day for the level, 0.02^2 and 0.05^2 per day for skew and curvature, plus a term in the spot move
K	The gain: the share of the total variance the prediction carries; 0 keeps the prediction, 1 takes the reading
h, SSR	Log forward move since the last reading; the spot-vol rule of slide 31, written SSR here because R is a variance
$\mathcal{T}_h$	The transport of line 4

WORKED EXAMPLE

Level: prediction $20.0\% \pm 0.30\%$, today's fit $20.4\% \pm 0.15\%$:
 $K = 0.30^2 / (0.30^2 + 0.15^2) = 0.80$, posterior $20.32\% \pm 0.13\%$. Curvature: prediction 0.10 ± 0.05 , fit 0.55 ± 0.30 (a stale strike puts a kink in it): $K = 0.027$, posterior 0.112 . The kink is barely admitted.

Where the reading's variance comes from

The variance of today's handles is computed from the fit itself: from how firmly the quotes, at their stated noise, pin the parameters, and from how badly the fit missed them.

TODAY'S FIT: PARAMETERS θ , m QUOTES, HANDLES $x(\theta) = (\sigma_0, s_0, \kappa_0)$

1 · STATED NOISE OF A QUOTE, IN VOL: HALF ITS SPREAD, FLOORED AT 1 BP, WIDER UNDER 30 DAYS

$$\delta_i = \max\left(\frac{1}{2}(\sigma_i^{\text{ask}} - \sigma_i^{\text{bid}}), 1 \text{ bp}\right) \cdot \max\left(1, \sqrt{30/n_d}\right)$$

2 · HOW FIRMLY THE QUOTES PIN THE PARAMETERS: THE JACOBIAN, EACH QUOTE ROW DIVIDED BY ITS NOISE

$$\mathcal{I}_\theta = J^\top J, \quad J_{ij} = \frac{1}{\delta_i} \frac{\partial r_i}{\partial \theta_j} \quad (\text{quote rows}), \quad \text{plus the penalty rows}$$

3 · FROM PARAMETERS TO HANDLES: THE HANDLE JACOBIAN G CARRIES THE UNCERTAINTY THROUGH

$$R_x = G \mathcal{I}_\theta^{-1} G^\top, \quad G = \frac{\partial x}{\partial \theta} \quad (3 \times (N + 1))$$

4 · INFLATE WHEN THE FIT MISSES THE QUOTES BY MORE THAN THEIR STATED NOISE

$$R = \rho R_x, \quad \rho = \text{clip}\left(\frac{\chi^2}{m - 3}, 1, 25\right), \quad \chi^2 = \sum_i \lambda_i \frac{r_i^2}{\delta_i^2}$$

r_i : residual of quote i in vol (slide 10), λ_i its weight (slide 12), both at the solution. J : one row per quote and per penalty, one column per parameter. $\mathcal{I}_\theta^{-1}$: regularised — eigenvalues under 10^{-10} of the largest are raised, so an unpinned direction reads as a large, finite variance. n_d : calendar days to expiry.

- **Reading the Jacobian:** $J^\top J$ says how much a unit move of each parameter would change the residuals, in units of the quotes' noise: large means pinned. G says how much each handle moves per parameter. $G \mathcal{I}^{-1} G^\top$ is then the variance of the handles that the quote noise implies.
- **Contradiction is noise:** a cluster the fit cannot match within its stated noise raises χ^2 , so ρ and R grow and the gain falls, with no threshold.
- **Overlay or active:** Overlay, the default, computes and draws the filtered handles; the fit is untouched. Active adds the prediction to the fit as three rows $(x_j(\theta) - m_j^-)^2 / P_j^-$, and the fit's own handles become the posterior.

WORKED EXAMPLE

Twenty quotes, each with 10 bp of stated noise, all reading the level: $\mathcal{I} \approx 20/\delta^2$, so the level's sd is $\delta/\sqrt{20} = 2.2$ bp. If the fit misses them by twice their noise, $\chi^2 \approx 80$ and $\rho \approx 80/17 = 4.7$: the sd becomes $2.2\sqrt{4.7} = 4.9$ bp and the gain falls.

IN THE APP

Options ▶ Kalman filter: R Jacobian (default) or factors; the p column of the filter table; the contaminated flag when the fit carried prior rows.

Series replay and independent calibration lanes

A series stores one ticker's chains at a sequence of instants. Several lanes calibrate the same frames in time order, each with its own settings and its own memory, so their differences come from the settings alone.

LANE ℓ OVER FRAMES $i = 1 \dots n$, FOR THE EXPIRY ON SCREEN

1 · MEMORY: A LANE'S PRIOR IS ITS OWN PREVIOUS FIT, TRANSPORTED

$$\text{prior}_i^\ell = \mathcal{T}_{h_i}(\text{fit}_{i-1}^\ell), \quad h_i = \log \frac{F_i}{F_{i-1}}$$

2 · FIT ERROR: MEAN OVER FRAMES OF THE FIT'S RMS, IN VOL BP

$$\overline{\text{rms}}^\ell = \frac{1}{n} \sum_{i=1}^n \text{rms}_i^\ell$$

3 · ROUGHNESS: MEAN FRAME-TO-FRAME MOVE OF THE ATM VOL

$$\text{rough}^\ell = \frac{1}{n-1} \sum_{i=2}^n |\sigma_0^\ell(i) - \sigma_0^\ell(i-1)|$$

4 · PULL: ATM DISTANCE TO THE FREE LANE AT THE SAME FRAME

$$\text{pull}_i^\ell = \sigma_0^\ell(i) - \sigma_0^{\text{free}}(i)$$

$\mathcal{T}_h$: the spot transport of slide 31. F_i : the expiry's forward at frame i . $\sigma_0^\ell(i)$: ATM vol of lane ℓ at frame i . A free lane has no memory; a filter lane also carries its own Kalman state, and $\zeta = (z - m^-) / \sqrt{P^- + R}$ has a spread near 1 when its variances are right.

OBJECT	HOLDS	PURPOSE
Frame	One instant: chain, spot and quote kind, harvested, live or imported	The same market inputs for every lane
Lane	A patch over the series' frozen base settings: model, prior mode, filter mode	Change one policy or model at a time
Lane memory	Its own prior and filter state, checkpointed at every frame	No lane feeds another; pause and resume are exact
Replay	Frames in time order; each lane runs the desk's own Calibrate on a detached state	No future information; the desk's fit
Readouts	Smile, surface, term; per lane rms, roughness, pull, filter ζ , fit time	Where and when the lanes diverge, and at what cost

- **Three questions:** fit to the quotes of each frame (rms), stability through time (roughness), and response to a real move. A prior or a filter buys stability at some rms; a smoother path alone does not prove a better estimate.
- **Presets:** LQD free, + prior (hybrid), + prior + filter (active), SVI-JW free, MCS free, LV free, LV + prior, and Current Options. A memory lane's first frame is a seed fit.

IN THE APP

Series lens (Alt+6): New series... with clock, ladder and lane presets, Estimate, then Start; the Lanes stage shows the metric chart and the evidence table (mean rms · roughness · pull · ζ · fit ms).

RECORDED, SPY REPLAY DAY, 10 SEPTEMBER

25 frames, 15 minutes apart; one-day expiry. Free: rms 4.66 bp, ATM roughness 18.4 bp per frame. + Prior (hybrid): rms 5.20 bp, roughness 17.5 bp, mean |pull| 3.6 bp. The overlay filter leaves the fit unchanged and adds its record: ATM ζ standard deviation 0.82.

Graph state: handle changes against transported priors

A node is one smile: one underlying at one expiry. The graph estimates how far today's market has moved each node's three ATM handles away from its saved prior.

NODE i : ITS SMILE $\sigma_i(k)$ IN LOG-MONEYNESS $k = \log(K/F)$

1 · THREE HANDLES y_i : LEVEL, SLOPE AND CURVATURE AT THE MONEY

$$y_i = \left(\sigma_i(0), \partial_k \sigma_i(0), \partial_k^2 \sigma_i(0)\right)$$

2 · LIT NODE: TODAY'S FIT MINUS THE TRANSPORTED PRIOR

$$d_i = y_i^{\text{fit}} - y_i^0$$

3 · EVERY NODE AFTER THE SOLVE: PRIOR PLUS INFERRED CHANGE

$$\hat{y}_i = y_i^0 + \hat{z}_i, \quad \hat{z}_i = 0 \text{ if no lit node is connected}$$

y_i^0 : the saved prior, moved to today's forward by the spot-vol rule; taken from this expiry of the active prior, else its nearest expiry, else today's fit, else a flat 20% smile. z_i : the unknown change at node i ; $\hat{z}_i$: its estimate.

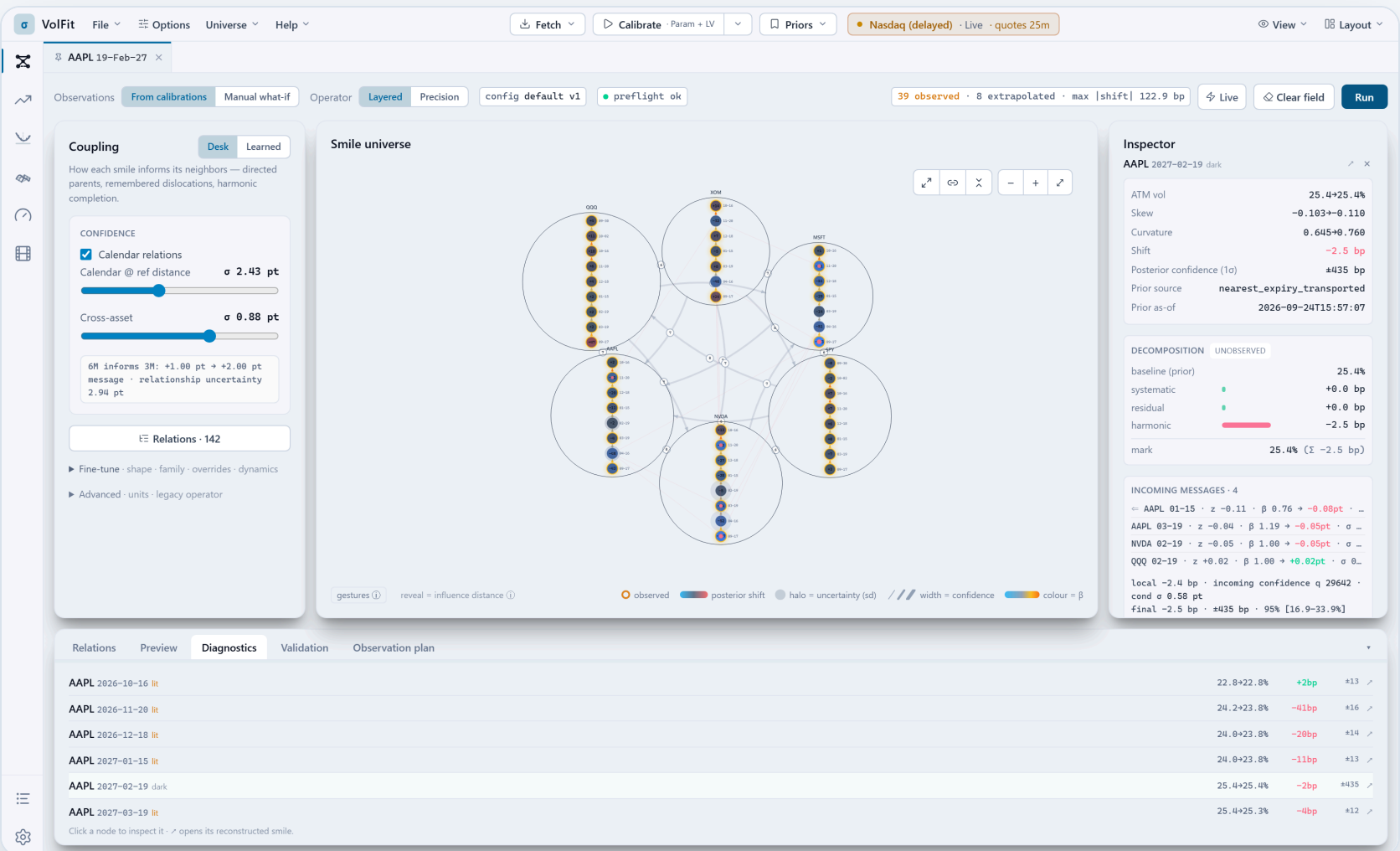
- **Why changes:** the prior keeps each smile's own shape. The graph moves only what the market moved; a dark node stays at its prior until lit data reach it.
- **Lit and dark:** a lit node's d_i is observed today. A dark node's z_i is inferred through the relations, and its band says how well.

RECORDED, 24 SEPTEMBER

Dark AAPL 19 Feb 2027, Layered Run. Baseline from the prior's nearest expiry, $y^0 = (25.4\%, -0.103, 0.645)$; result $\hat{y} = (25.4\%, -0.110, 0.760)$, so $\hat{z} = (-2.5 \text{ bp}, -0.007, +0.115)$. The lit AAPL expiries moved between +2 and -41 bp.

IN THE APP

Graph lens (Alt+1): Run, then click a dark node. The Inspector shows prior to posterior per handle, the Shift, the Posterior confidence (1σ) and the Prior source.



24 September, after a Layered Run: 39 lit nodes (filled) and 8 dark (outlined) on six underlyings, SPY, NVDA, AAPL, QQQ, XOM and MSFT. Arrows are calendar and cross-asset relations; the largest shift is 122.9 bp.

Three graph operators and what each assumes

The operator is the rule that carries lit changes to dark nodes. Layered is the default, Precision the second choice, Smooth field the legacy option under Advanced.

EACH OPERATOR: A QUADRATIC PENALTY $E(z)$ ON THE CHANGES, MINIMISED GIVEN THE LIT DATA

1 · SMOOTH FIELD: STAY SMALL, AND CLOSE TO THE NEIGHBOURS' WEIGHTED AVERAGE

$$E_{\text{sm}}(z) = \kappa \sum_i z_i^2 + \eta \sum_i \pi_i \left(z_i - \sum_j K_{ij} z_j \right)^2$$

2 · PRECISION: ONE TERM PER RELATION $z_i \approx \beta_{ij} z_j$, HELD WITH PRECISION p_{ij}

$$E_{\text{msg}}(z) = \sum_{j \rightarrow i} p_{ij} \left(z_i - \beta_{ij} z_j \right)^2 = z^\top Q z$$

3 · LAYERED, RECIPROCAL PART: LIT NODES FIXED AT THEIR OBSERVED CHANGES d_{lit} , ZERO GRADIENT IN THE FREE NODES

$$Q_{\text{free,free}} \hat{z}_{\text{free}} = - Q_{\text{free,lit}} d_{\text{lit}}$$

z : the changes at all nodes. κ : pull toward zero; η : reach. K_{ij} : trust weights, rows summing to 1; π_i : node weight, the stationary mass of K . p_{ij} , β_{ij} : precision and amplitude of relation $j \rightarrow i$. d_{lit} : the observed changes $d_i = y_i^{\text{fit}} - y_i^0$ of the lit nodes (slide 37), stacked; $\hat{z}_{\text{free}}$: the changes solved for at the free nodes, the ones to fill; $Q_{\text{free,lit}}$: the block of Q linking them to the lit nodes.

- **Probability reading:** E is minus twice the log-density of a Gaussian prior on z . Adding the lit data and minimising gives the posterior mean; the curvature of E gives the bands.
- **Beyond line 3:** Layered also carries one-way arcs with a remembered residual, four slides on.

OPERATOR	ASSUMES	FAR FROM THE LIT NODE	DIRECTION · MEMORY
Smooth field · legacy	related smiles move alike	the move fades with distance	none · none
Precision	each relation is a stated rule: β and p	full β amplitude at Desk strength; the band widens	both ways · none
Layered · default	the same rules, plus one-way arcs where set	full β amplitude; fresh lit nodes stay fixed	one way on directed arcs · each node's residual

WORKED EXAMPLE

Ladder 3M, 6M, 1Y of one ticker, only 6M lit at +1 vol point. Precision and Layered: 3M +2.00, 1Y +0.50, because $\beta = T_{\text{informer}}/T_{\text{receiver}}$, whatever the precision. Smooth field at its defaults: +0.89 at both ends, since its calendar edges carry $\beta = 1$.

One relation: amplitude and confidence

A relation states how much a receiver moves when its informer moves, and how firmly. The two answers are separate numbers: β and σ .

RELATION $j \rightarrow i$: INFORMER j , RECEIVER i , IN THE RECEIVER'S UNITS

1 · THE STATEMENT: AMPLITUDE β , UNEXPLAINED PART OF SIZE σ

$$z_i = \beta_{ij} z_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2), \quad p_{ij} = \frac{1}{\sigma_{ij}^2}$$

2 · DEFAULT CALENDAR AMPLITUDE: THE RATIO OF MATURITIES

$$\beta_{ij} = \left(\frac{T_j}{T_i}\right)^{\alpha_T}, \quad \alpha_T = 1$$

3 · EACH RELATION ADDS ONE RANK-ONE TERM TO THE PRECISION MATRIX

$$Q_{\text{msg}} = \sum_{j \rightarrow i} p_{ij} u_{ij} u_{ij}^\top, \quad u_{ij} = e_i - \beta_{ij} e_j$$

4 · WHAT THE RECEIVER HEARS, IF ITS INFORMERS WERE KNOWN

$$z_i \mid z_{\text{informers}} \sim \mathcal{N}\left(\sum_j \frac{p_{ij}}{q_i} \beta_{ij} z_j, \frac{1}{q_i}\right)$$

- **Why the ratio:** an ATM change z moves total variance $\sigma_{\text{ATM}}^2 T$ by about $2\sigma_{\text{ATM}} T z$. The same variance change at two maturities, with similar vols, gives $z_i = (T_j/T_i) z_j$.
- **Two dials:** one known informer gives the mean $\beta_{ij} z_j$ at any p_{ij} ; p_{ij} sets only the band.

IN THE APP

Coupling pane: the live line '6M informs 3M: +1.00 pt → +2.00 pt message · relationship uncertainty 2.94 pt' follows the dials.

SYMBOL	MEANING
z_i, z_j	ATM change at receiver i and informer j ; 0.01 is one vol point
β_{ij}	Amplitude: receiver move per unit informer move
$\varepsilon_{ij}, \sigma_{ij}$	The part of the move the relation leaves unexplained, and its standard deviation
p_{ij}	Relation precision $1/\sigma_{ij}^2$, per vol ²
T_i, α_T	Years to expiry (days / 365.25); shape exponent, 1 by default
e_i, u_{ij}	Unit vector at node i ; the factor's direction $e_i - \beta_{ij} e_j$
Q_{msg}	The relations' precision matrix: $z^\top Q_{\text{msg}} z = \sum p_{ij} (z_i - \beta_{ij} z_j)^2$
q_i	Incoming precision: $\sum_j p_{ij}$ over the informers of i

WORKED EXAMPLE

6M lit at +1 point, 3M and 1Y dark. 3M: $\beta = 0.5/0.25 = 2$, so +2.00 with $1\sigma = 2.94$ points. 1Y is the informer of the 6M–1Y relation (the shorter expiry receives), so it is read backwards: $\beta = 1/2$, $\sigma = 3.14/2$, so +0.50 ± 1.57. The two σ are the next slide's calendar rule, $\sigma = \sqrt{(0.97 + \sqrt{\text{gap}})/1700}$: 2.94 at a 0.25-year gap, 3.14 at 0.5. Multiplying every p by 100 keeps the moves and divides the bands by 10.

Calendar and cross-asset relation controls

Two sliders set the default confidence of every automatic relation. Clicking an arrow opens the relation card, which edits that one relation: its confidence σ , its amplitude β and, in Layered, its direction.

HOW A RELATION'S DEFAULT CONFIDENCE IS SET, AND HOW IT READS FROM ITS OTHER END

1 · CALENDAR RELATION BETWEEN EXPIRIES T_i AND T_j : CONFIDENCE FALLS SLOWLY WITH THE GAP

$$p_{ij} = \frac{p_0}{\varepsilon_T + \sqrt{|T_i - T_j|}}, \quad \sigma_{ij} = \frac{1}{\sqrt{p_{ij}}}$$

2 · THE SAME RELATION READ FROM THE OTHER END: SAME PENALTY, THE OTHER NODE'S UNITS

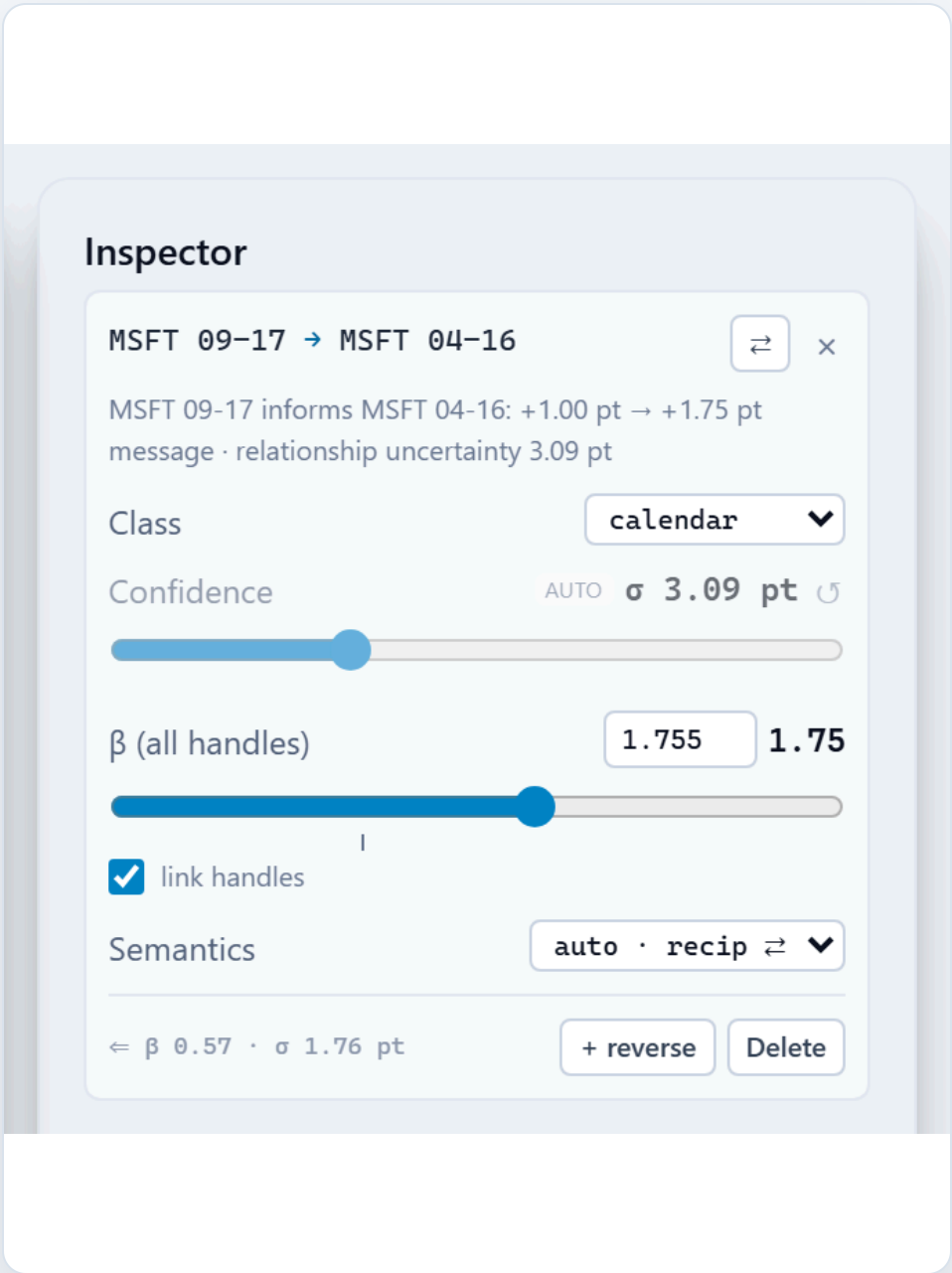
$$p(z_i - \beta z_j)^2 = p\beta^2\left(z_j - \frac{z_i}{\beta}\right)^2 \Rightarrow \beta_{\text{rev}} = \frac{1}{\beta}, \quad \sigma_{\text{rev}} = \frac{\sigma}{|\beta|}$$

$p_0 = 1700$ per vol^2 and $\varepsilon_T = 0.97 \sqrt{\text{years}}$, both fitted on stored day-over-day moves of adjacent expiries. So $\sigma = 2.55$ points at a one-week gap, 2.94 at three months, 3.40 at one year; the shorter expiry is the receiver. Cross-asset relations, same expiry: $\beta = 1$ and one constant $p_{\times} = 1.3 \times 10^4$, $\sigma = 0.88$ point.

- **Two sliders:** Calendar moves p_0 ; it reads 2.43 points, the σ at a reference gap where $\varepsilon_T + \sqrt{\text{gap}} = 1$. Cross-asset moves $p_{\times}$.
- **One card:** confidence: AUTO from line 1, or a value; amplitude β : the maturity ratio by default; semantics (Layered): reciprocal, as every automatic relation is, or directed, a one-way arc. Its last line is line 2, the reverse reading.
- **Editing:** every change stages a draft; Live previews it without recording; Run solves and records it; Apply activates it, Discard reverts.

WORKED NUMBERS, THE MSFT CARD

Gap 154 days = 0.422 years, $\sqrt{0.422} = 0.649$: $p = 1700 / (0.97 + 0.649) = 1050$, $\sigma = 1 / \sqrt{1050} = 3.09$ points. +1 point in September 2027 predicts +1.755 in April 2027, give or take 3.09; read backwards, +1 in April predicts +0.57 in September, give or take 1.76.



24 September, MSFT relation card: 17 Sep 2027 informs 16 Apr 2027. $\beta = 358/204 = 1.755$, the ratio of days to expiry, and $\sigma = 3.09$ points from line 1; read backwards (bottom left), $\beta = 0.57$ and $\sigma = 1.76$ points.

The joint posterior and shared information

All relations and all lit changes are solved together, one linear system per handle. A source reached by two routes then counts once.

THE PRECISION OPERATOR'S SOLVE, ONE HANDLE AT A TIME

1 · POSTERIOR PRECISION: RELATIONS, ANCHORS, LIT OBSERVATIONS

$$Q^+ = Q_{\text{msg}} + \text{diag}(\kappa) + \sum_{s \in \text{lit}} r_s e_s e_s^\top$$

2 · POSTERIOR MEAN AND COVARIANCE

$$Q^+ \hat{z} = \sum_{s \in \text{lit}} r_s d_s e_s, \quad \Sigma^+ = (Q^+)^{-1}$$

3 · A LIT CHANGE IS UNCERTAIN TWICE: TODAY'S FIT AND THE BASELINE

$$\frac{1}{r_s} = \frac{1}{r_s^{\text{fit}}} + \frac{1}{p_s^0}$$

4 · REPORTED 1σ: THE MARGINAL; A DARK NODE ADDS ITS BASELINE ONCE

$$\text{sd}_i = \sqrt{\Sigma_{ii}^+} \text{ (lit),} \quad \text{sd}_i = \sqrt{\Sigma_{ii}^+ + 1/p_i^0} \text{ (dark)}$$

- Which number is the confidence: the marginal Σ_{ii}^+ . The incoming q_i treats the informers as exact: dark AAPL Feb 2027 on 24 September had $q = 29,642$, a conditional $1/\sqrt{q} = 0.58$ point, and a final ± 4.35 points.
- No lit path: a group of nodes with no lit node is not solved: $\hat{z} = 0$, the prior, with a band of about 3 points on the level.

SYMBOL MEANING

$\hat{z}, \Sigma^+$	Posterior mean of the changes, and their covariance
Q_{msg}	The relations' precision matrix, previous slides
κ_i	Pull toward zero, $p_{\text{max}}(1 - \rho)/\rho$, p_{max} the node's firmest relation; 0 at Desk
e_s, d_s	Unit vector and observed change at lit node s
r_s	Precision of d_s : fit and baseline combined
r_s^{fit}	$1/\text{rms}^2$, cut for few quotes, wide spreads, old data
p_i^0	Baseline precision: source tier, cut by age and transport; $\times 0.25$ if dark
q_i	Incoming precision $\sum_j p_{ij}$: conditional, not the band

WORKED EXAMPLE

A lit with variance $1/p$; relations A–B, A–C, B–C, all $\beta = 1$ and precision p . The routes to C, direct ($1/p$) and via B ($2/p$), combine to $2/(3p)$; A's own $1/p$ enters once: $\text{Var}(z_C) = 5/(3p)$. Independent routes would count A twice: $6/(5p)$, 28% too small; at $\sigma = 1$ point, 1.10 against 1.29.

Layered propagation and residual memory

Layered runs in two layers: one-way arcs first, where a target's mark is its sources' prediction plus its own remembered dislocation; then the reciprocal completion.

LAYER 1, THE DIRECTED ARCS; THEN LAYER 2, THE RECIPROCAL COMPLETION

1 · LAYER 1, DIRECTED ARC $j \rightarrow i$: THE SOURCES' PREDICTION s_i PLUS THE TARGET'S RESIDUAL u_i

$$m_i = \underbrace{\sum_j \frac{p_{ij}}{q_i} \beta_{ij} m_j}_{s_i} + u_i$$

2 · THE RESIDUAL: SET WHEN THE TARGET PRINTS, THEN HALVED EVERY H DAYS

$$u_i \leftarrow d_i - s_i, \quad u_{i,t+\Delta} = 2^{-\Delta/H} u_{i,t}$$

3 · LAYER 2: FRESH LIT NODES FIXED AT d_{lit} ; THE LAYER-1 MARKS AS SOFT ANCHORS

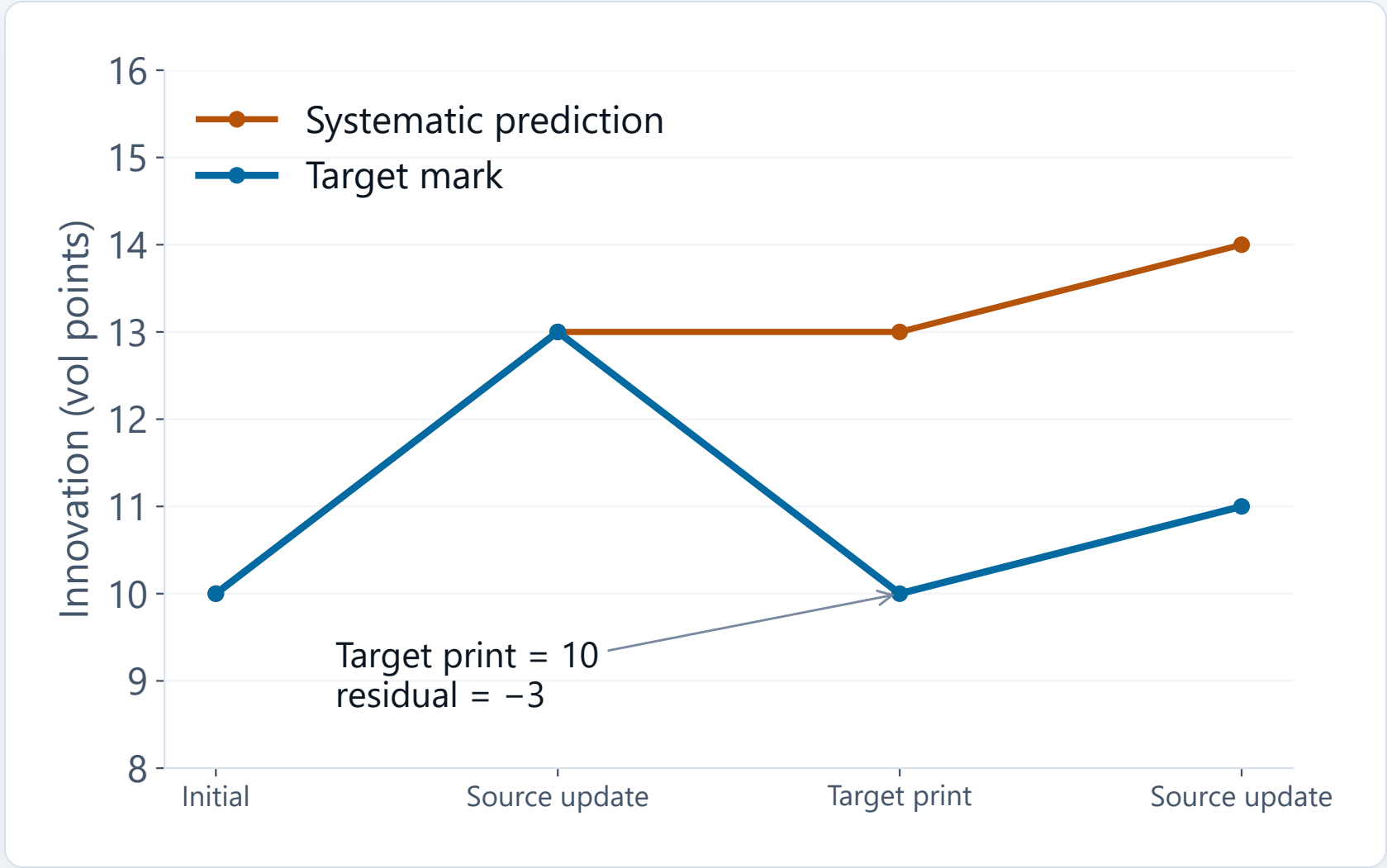
$$\left(Q_{\text{free,free}} + \text{diag } \frac{1}{V}\right) \hat{z}_{\text{free}} = -Q_{\text{free,lit}} d_{\text{lit}} + \frac{m}{V}$$

m_i : mark of node i ; s_i : systematic part, the precision-weighted average of the sources' messages $\beta_{ij} m_j$. d_i : the change the target printed; u_i : what it left unexplained; Δ : days since it; H : half-life, never by default. d_{lit} : the fresh lit nodes' changes; V_i : variance of a layer-1 mark.

- **Why one-way:** an index informs its names; a name printing cheap must not mark the index down. The name's own dislocation lives in its residual.
- **Memory in the default:** each Layered Run stores every fresh lit node's change. If the node is dark at a later Run, that change anchors it, unchanged while H is never.

WORKED EXAMPLE

$H = 5$ days, the source moving one day after the print: $u = -3 \times 2^{-1/5} = -2.61$, so the target marks $14 - 2.61 = +11.39$; after 5 days $u = -1.50$. With H never, u stays at -3 until the next print.



Illustrative, $\beta = 1$, H never. The target follows its source to +13, then prints +10: residual -3 . The source moves to +14; the target marks $14 - 3 = +11$. The source's path never reacts to the target's print.

Reconstructing and displaying an inferred smile

The graph returns three numbers per node: ATM level, skew and curvature. The node's own earlier shape supplies the rest, so the rebuilt curve is always a valid LQD law.

NODE i AFTER A RUN; HANDLES $y = (\sigma_0, s_0, \kappa_0)$: ATM VOLATILITY, SKEW, CURVATURE

1 · TARGET HANDLES: THE BASELINE PLUS THE GRAPH'S MOVE
$$\hat{y}_i = y_i^0 + \hat{z}_i$$

2 · RETARGET THE BASE SLICE: SOLVE FOR THREE MOVES, SHAPE MODES HELD
$$y(\theta^* + U\delta + V\xi^*) = \hat{y}_i$$

3 · BAND $\hat{\sigma}(k) \pm 1.96 \text{sd}(k)$, FIRST ORDER IN THE HANDLE SDS
$$\text{sd}(k)^2 = \sum_{j=1}^3 \left(\frac{\partial \sigma(k)}{\partial y_j} \right)^2 \text{sd}_j^2$$

y_i^0 : transported baseline; $\hat{z}_i$: the graph's move. θ^* : LQD parameters of the base slice (the node's latest fit, else its transported prior). $U\delta$: three moves that change the handles, sizes δ by Newton; $V\xi^*$: all other shape modes, fixed. sd_j : marginal posterior sd of handle j . σ_I : the ticker's recent unexplained ATM move. κ_0 is curvature, not the graph anchor.

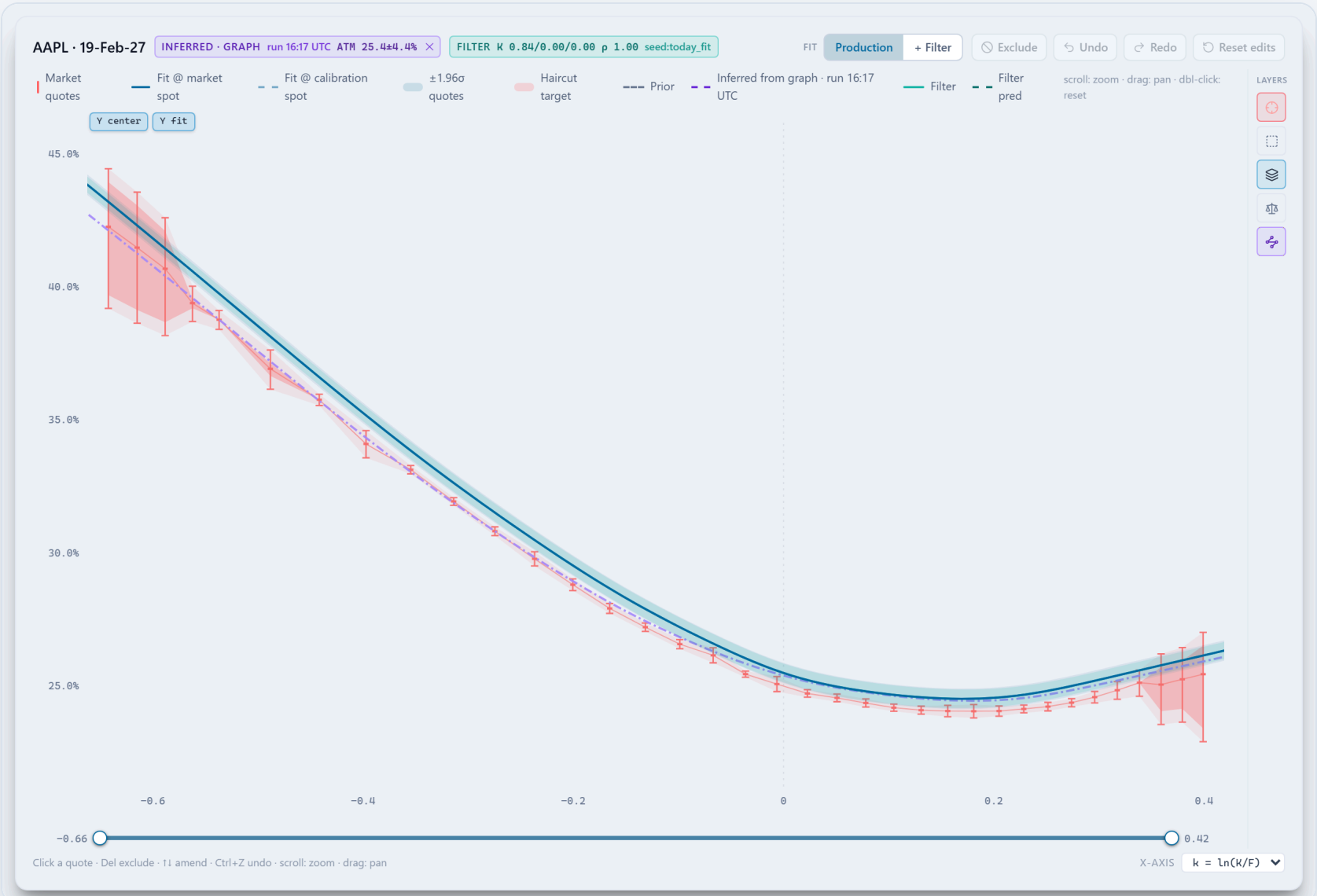
- **Always a valid law:** only LQD parameters move. An SVI or MCS desk model is then fitted to this LQD target.
- **Never a fit:** tagged graph: violet dash-dot, transported with the spot, never committed, never a prior.

WORKED NUMBERS, 24 SEPTEMBER CAPTURE

AAPL February 2027, dark: baseline 25.4%, graph move −2.5 bp. Its ATM sd is the floor $\sqrt{0.30} \sigma_I = \sqrt{0.30} \times 7.95 = 4.35$ points, from AAPL moves stored in July. 95%: $25.4 \pm 1.96 \times 4.35$, so 16.9 to 33.9%.

IN THE APP

After a Run, a dark node's smile: the INFERRED · GRAPH badge and the violet curve. Opened from the Graph lens it adds the 95% band and RMS · in-band · ζ against the node's quotes.



AAPL 19 February 2027, dark, 24 September. Violet dash-dot: the smile inferred by the 16:17 UTC Run. Blue: the node's earlier fit, now stale. Red: its current quotes, not used by the Run. Badge: ATM $25.4 \pm 4.4\%$, one sd.

Validation against a transported baseline

Hide one lit node's calibration, infer it from everything else, and compare. The baseline to beat is the transported prior, not a flat surface.

LIT NODE i WITHHELD; ATM VOLATILITY $\sigma_{0,i}$; n SCORED NODES

1 · ERRORS OF THE GRAPH AND OF THE BASELINE

$$e_i = \hat{\sigma}_{0,i}^{(-i)} - \sigma_{0,i}^{\text{fit}}, \quad e_i^0 = \sigma_{0,i}^0 - \sigma_{0,i}^{\text{fit}}$$

2 · STANDARDISED RESIDUAL: THE ERROR IN UNITS OF ITS STATED UNCERTAINTY

$$\zeta_i = \frac{\sigma_{0,i}^{\text{fit}} - \hat{\sigma}_{0,i}^{(-i)}}{\sqrt{V_i^{(-i)} + 1/r_i}}$$

3 · COVERAGE: SHARE OF NODES INSIDE THE 80% AND 95% BANDS

$$\text{cov}_p = \frac{1}{n} \# \{i : |\zeta_i| \leq z_p\}, \quad z_{80} = 1.28, \quad z_{95} = 1.96$$

$\sigma_{0,i}^{\text{fit}}$: calibrated ATM volatility of node i . $\hat{\sigma}_{0,i}^{(-i)}$, $V_i^{(-i)}$: posterior mean and variance there, with it withheld. $\sigma_{0,i}^0$: its transported prior. r_i : precision of its own ATM estimate. $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i e_i^2}$.

- **What is withheld:** only node i 's current calibration. Its prior stays, and on this desk the priors were saved the same afternoon.
- **Reading ζ :** a standard deviation near 1 means bands of the right width; below 1, too wide; above 1, too narrow. Coverage should read near 80% and 95%.

WORKED NUMBERS, 24 SEPTEMBER TABLE

Smooth field: RMSE 36.7 bp, 2.0 better than the baseline's 38.7, but ζ std 0.11 and 39 of 39 inside 80%: bands about nine times too wide. Messages: 38.5 bp; ζ std 0.72; 35 of 39 inside 80%, 37 inside 95%.

IN THE APP

Graph drawer ▶ Validation ▶ Compare operators (LOO): one solve per held-out node and operator, so start it while talking.

Relations Preview Diagnostics Validation Observation plan

⚠ Compare operators (LOO) [offline benchmark artifact](#)

	Transported prior	Smooth field	Messages
RMSE (bp)	38.7	36.7	38.5
ζ mean	—	−0.02	0.06
ζ std	—	0.11	0.72
cov 80	—	100%	90%
cov 95	—	100%	95%
n scored	39	39	39

Same day, same universe, same knobs — only the operator differs. The transported-prior column has no calibrated uncertainty ($\zeta/\text{cov n/a}$). The multi-regime verdict lives in the offline artifact.

24 September, Validation ▶ Compare operators (LOO), 39 held-out nodes. Same day, universe and settings; only the operator differs. The transported prior carries no stated uncertainty here, so it has no ζ or coverage.

Choosing the next observation

Before a quote exists, the posterior covariance says how much it would reduce uncertainty, here and at every correlated node. The plan ranks nodes by that.

CANDIDATE c , QUOTE NOISE VARIANCE $1/r_c$; Σ : POSTERIOR COVARIANCE OF ATM VOLATILITY

1 · COVARIANCE AFTER THE QUOTE: A RANK-ONE UPDATE

$$\Sigma' = \Sigma - \frac{\Sigma_{\cdot c} \Sigma_{c \cdot}}{\Sigma_{cc} + 1/r_c}$$

2 · VARIANCE REMOVED AT EACH NODE i

$$\Delta V_i = \frac{\Sigma_{ic}^2}{\Sigma_{cc} + 1/r_c}$$

3 · SCORE: THE REMOVED VARIANCE, SUMMED WITH EXPOSURE WEIGHTS

$$G(c) = \sum_i w_i \Delta V_i$$

$\Sigma_{\cdot c}$: column c of Σ . r_c : precision the quote would carry, from the candidate's own chain, else the median lit node's. w_i : exposure weight, 1 per ticker by default. The plan shows $G(c)/\sum_i w_i \Sigma_{ii}$ in % of σ^2 ; the idiosyncratic floor is left out.

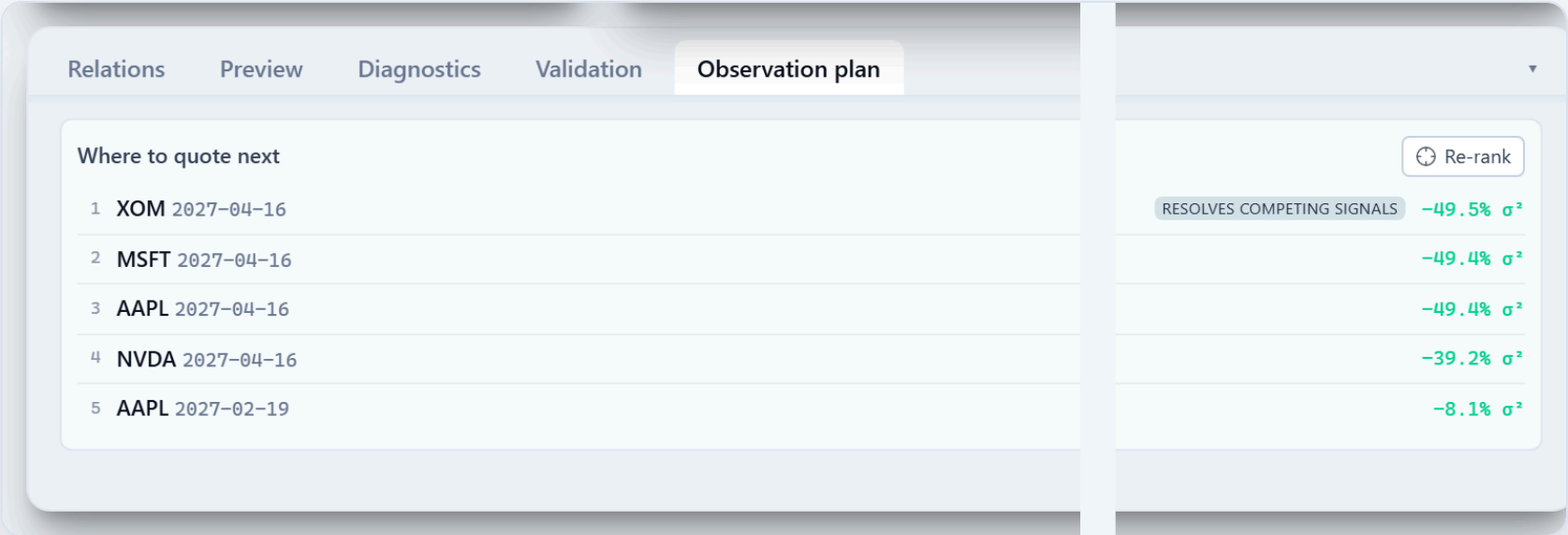
- **No quote value needed:** the update uses only the quote's precision. What the quote says moves the means once it arrives.
- **Why a node ranks high:** uncertain itself and correlated with many others, like the four dark April 2027 nodes, tied by same-expiry relations.

ILLUSTRATIVE CALCULATION

Nodes c and j , sd 60 and 50 bp, correlation 0.8: $\Sigma_{jc} = 2400 \text{ bp}^2$. Quote c at sd 15 bp: denominator $3600 + 225 = 3825$. At c , sd $60 \rightarrow 14.6$ bp; at j , $\Delta V = 2400^2/3825 = 1506$, sd $50 \rightarrow 31.5$ bp. 80% of their variance gone.

IN THE APP

Graph drawer ▶ Observation plan ▶ Re-rank: hover a row for sd before → after and the nodes it also shrinks; click to open its smile.



24 September, Observation plan ▶ Rank: the five best places to quote next. The April 2027 nodes lead, three of them tied near 49.4%; XOM's incoming messages disagreed in sign. AAPL February 2027, between two lit expiries, would remove 8.1%.

Where calibration time is spent

One SPY ticker with nine expiries: the smiles take a third of a second, the local-volatility surface about three seconds. Warm-starting the surface saves optimiser evaluations: 20 instead of 66.

RECORDED 23 SEPTEMBER · SPY, 9 EXPIRIES, 875 FIT QUOTES · ONE PROCESS, QUOTES ALREADY PREPARED

1 · ONE TICKER'S CALIBRATE JOB: THE SMILES, THEN THE SURFACE

$$T_{\text{job}} \approx T_{\text{LQD}} + T_{\text{LV}}$$

2 · LOCAL-VOLATILITY STAGE: FIXED WORK PLUS OPTIMISER EVALUATIONS

$$T_{\text{LV}} \approx T_{\text{fix}} + n_{\text{eval}} t_{\text{eval}}$$

3 · THE TWO RECORDED RUNS, COLD AND WARM PARAMETER START (SECONDS)

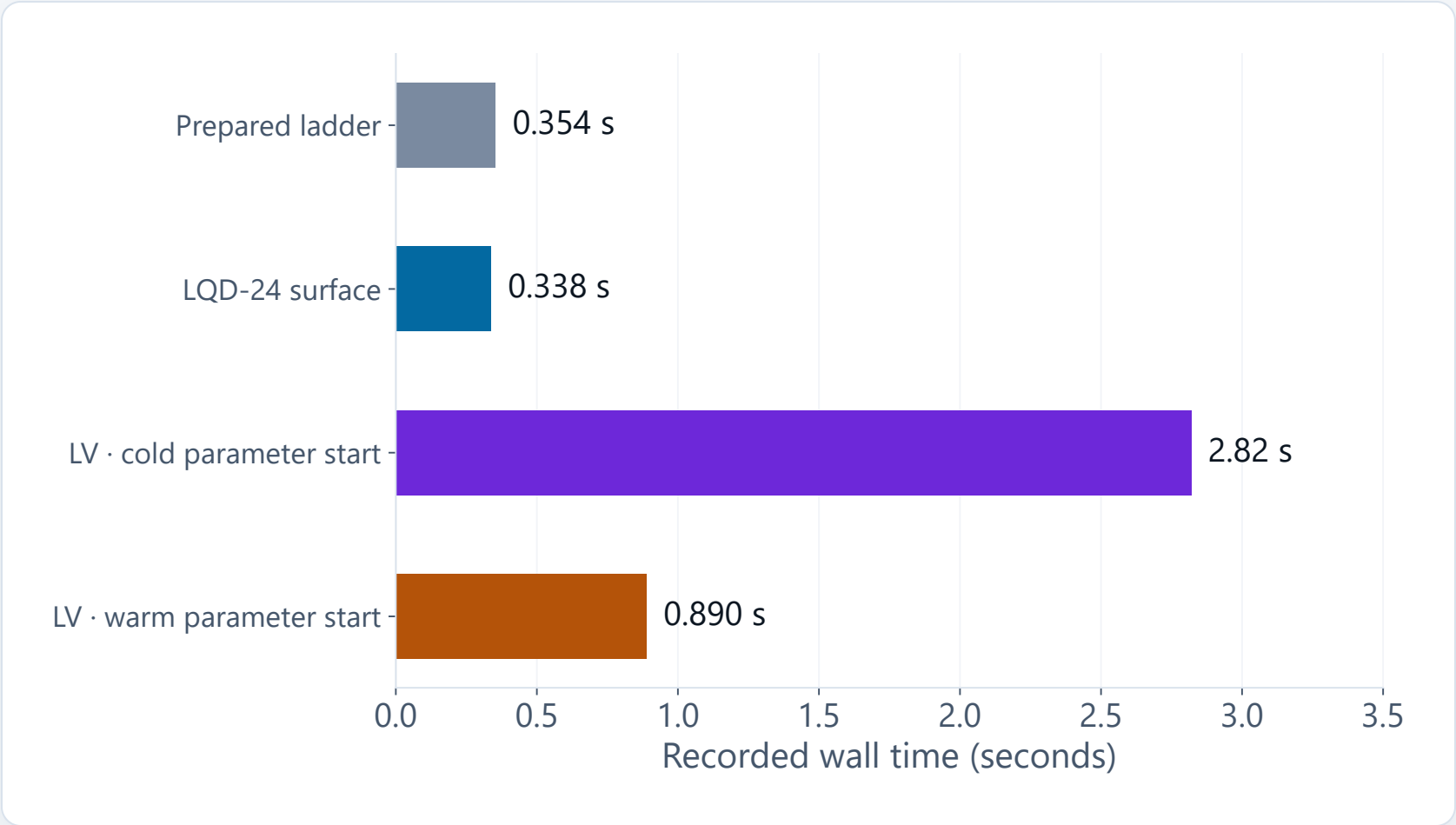
$$0.3 + 66 \times 0.038 \approx 2.8, \qquad 0.3 + 20 \times 0.030 \approx 0.9$$

T_{LQD} : nine LQD-24 smile fits with the calendar screen. T_{LV} : the local-variance surface, 253 vertices on a 368×116 pricing lattice. n_{eval} : optimiser evaluations, each a forward Dupire march with its sensitivities; t_{eval} : the cost of one. T_{fix} : work outside the optimiser loop: seed, three post-fit reprices, response.

- **Warm start:** the previous surface seeds the fit: 20 evaluations instead of 66, same result, 3.61 bp in the fitting operator and 5.12 bp on the converged reprice.
- **Cached work:** prepared quotes (forwards, exercise premium, inversions: 0.354 s for the ladder) are reused when only fit settings change; compiled kernels load from disk.
- **Parallelism:** the pool fits different tickers at once. One surface is one task: pooled 2.90 s against 2.82 s serial.

RECORDED 23 SEPTEMBER

Calibrate job 3.34 s, measured as one job; its two fit stages timed alone add to $0.34 + 2.82 = 3.16$ s. Before it: live fetch 1.2 to 1.4 s, quote preparation 0.35 s. One smile alone, after the fetch: about 50 ms.



Recorded 23 September after the US close: SPY Massive NBBO, mid target, i7-12700H laptop, one worker. Medians of three runs (the warm LV bar: one run). Cold: seeded from the LQD fits; warm: seeded from the previous surface.

Reading fit quality and publication state

The Quality lens reads cached fits only; it never refits. A lit node is ready when every gate passes; other diagnostics are shown but do not block.

QUOTES i OF ONE NODE, WEIGHTS λ_i , TARGET BAND $[\sigma_i^{\text{lo}}, \sigma_i^{\text{hi}}]$ OR MID σ_i^{mid}

1 · ERROR OF QUOTE i AGAINST ITS TARGET BAND: ZERO INSIDE

$$e_i = \left(\sigma(k_i) - \sigma_i^{\text{hi}}\right)^+ + \left(\sigma_i^{\text{lo}} - \sigma(k_i)\right)^+$$

2 · NODE RMS ON THE FIT'S OWN WEIGHTS

$$\text{RMS} = \left(\sum_i \lambda_i e_i^2 / \sum_i \lambda_i\right)^{1/2}$$

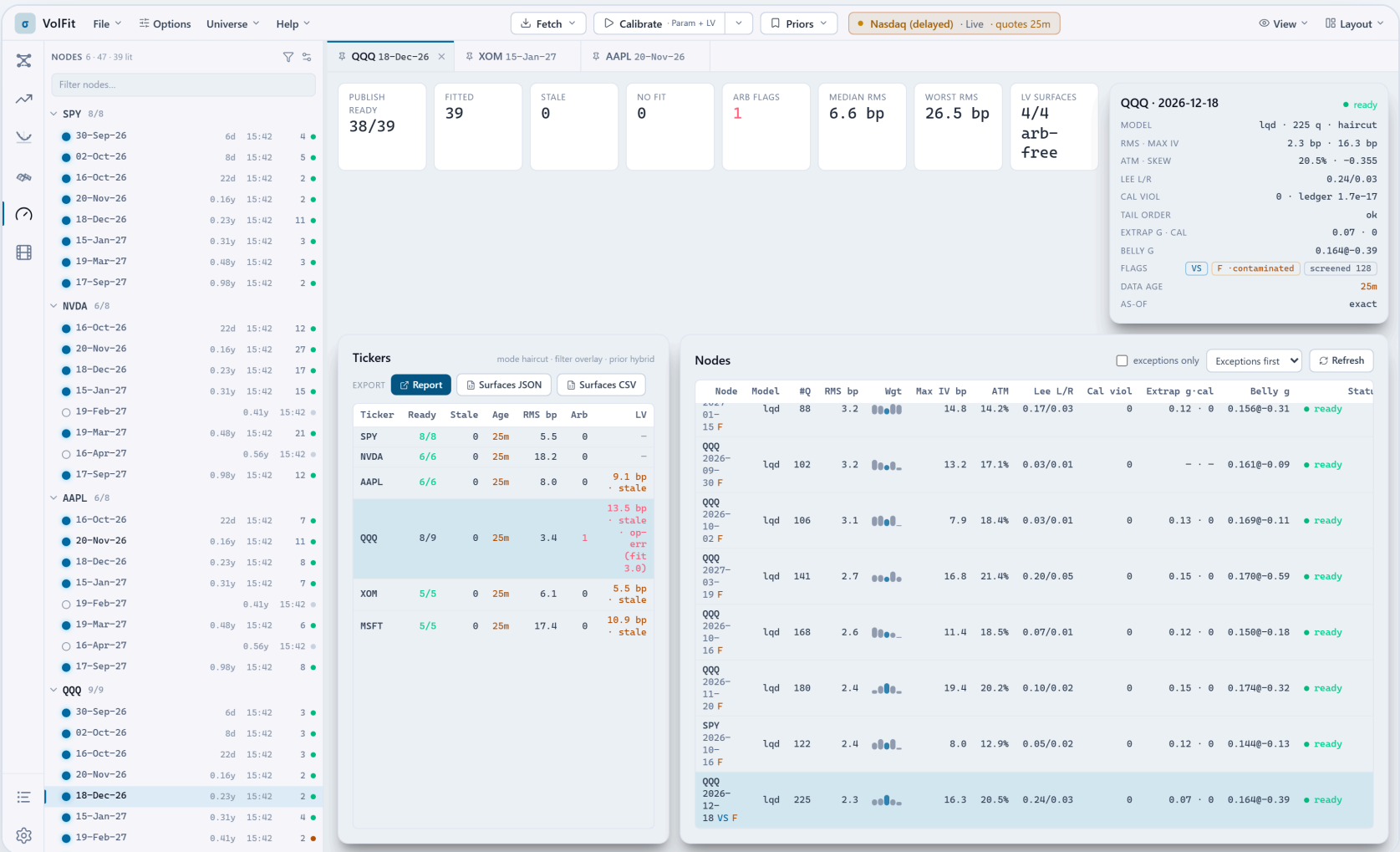
$\sigma(k_i)$: displayed model at quote i ; with a mid target, $e_i = \sigma(k_i) - \sigma_i^{\text{mid}}$. G : the ledger of slide 23, far minus near expiry at equal rank. g : Durrleman's function, $g \geq 0$ is a non-negative density. Calendar and butterfly failures also block an export.

GATE	READY REQUIRES	QQQ 18-DEC-26
Freshness	fit on current inputs; data under 120 min	current; 25 min
Fit error	$\text{RMS} \leq 50 \text{ bp}$	2.3 bp, max 16.3
Wing slopes	Lee $\beta_L, \beta_R \leq 2$	0.24, 0.03
Calendar	$\min_u (G_{\text{far}} - G_{\text{near}}) \geq -10^{-6}$	1.7×10^{-17}
Butterfly	$g \geq 0$ on the quoted range	min 0.164 at $k = -0.39$

RECORDED 24 SEPTEMBER

38 of 39 lit nodes ready. The exception, QQQ February 2027 (RMS 2.2 bp), fails the calendar certificate against January: minimum gap -0.1 bp of the forward at $k = -1.70$.

IN THE APP Quality lens (Alt+5): tiles, per-ticker rollup, node table with Exceptions first; a node card's certificates and data age.



24 September, Quality lens: publish ready 38/39, arbitrage flags 1 (QQQ), median RMS 6.6 bp, worst 26.5 bp (NVDA November 2026), LV surfaces 4/4 arbitrage-free. Card: QQQ December 2026.

A worked sequence on the 24 September desk

One liquid node, QQQ December 2026, and one dark node, AAPL February 2027, followed from quotes to the publish check. Every value is read off the archived captures.

STEP	WHERE IN THE APP	WHAT TO READ ON 24 SEPTEMBER
1 · Market input	Parametric ▶ Table, QQQ 18-Dec-26	chain 15:42:09 UTC; spot 736.05, forward 742.89, $T = 0.233$; 225 strikes
2 · One smile	Fit diagnostics card	LQD $N = 24$, haircut target: ATM 20.5%, skew -0.355 , curvature 4.01; $A_L = 0.628$, $A_R = 0.056$
3 · Expiry ladder	Stacked IV; Quality card	$w = \sigma^2 T$ for 9 expiries, chip 1 cal. cross; December against November: gap $1.7 \times 10^{-17} \geq 0$
4 · Spot move	Spot move card; Options ▶ Dynamics	SSR $R = 1.5$: the fit stays, the curve is transported (strip below)
5 · Earlier information	Fit switch; FILTER badge	Production against Free; filter gains K 0.61, 0.11, 0.00 for ATM, skew, curvature
6 · Dark node	Graph lens ▶ Inspector, AAPL 19-Feb-27	baseline 25.4% (nearest-expiry prior), move -2.5 bp, ± 435 bp (1 sd); violet curve on its smile
7 · Score and plan	Graph drawer ▶ Validation, Observation plan	39 held out: 38.7, 36.7, 38.5 bp; next quote XOM 16-Apr-27, -49.5% of σ^2
8 · Publish check	Quality lens (Alt+5)	38 of 39 ready; QQQ 19-Feb-27 fails the calendar certificate

WORKED EXAMPLE, STEP 4

QQQ falls 1%: $h = \log(F_1/F_0) = -0.01$. First-order ATM change $R s_0 h = 1.5 \times (-0.355) \times (-0.01) = +53$ bp, with no refit. Sticky strike, $R = 1$, would give +35.5 bp.

IN THE APP

Open each node by link: </?node=QQQ|2026-12-18&activity=parametric>, then </?node=AAPL|2027-02-19&activity=graph>.

CLOSE

One calculation, read in dependency order

Every number on the desk is observed, fitted, carried or inferred. Each step between them adds one assumption, and the app shows it beside the number.

- **Observed:** quotes brought to one spot and time, the forward and discount from parity, the early-exercise premium removed.
- **Fitted:** one smile per expiry, LQD, SVI or MCS, or one local-variance surface, read with its target, weights, tails and calendar order.
- **Carried:** a fit moved with the spot by the spot–vol rule R , a prior where quotes are thin, a filter across time.
- **Inferred:** dark nodes from the graph: baseline plus a relation-weighted move, rebuilt as a smile with its band; never a fit, never a prior.
- **Dated:** timings from the 23 September SPY run; captures and scores from the 24 September session.